

Exact neutrino oscillation probabilities: one construction, its extension to four flavours, and the limits of the closed form

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Abstract

The exact methods used to compute neutrino oscillation probabilities in a constant medium—the Ohlsson–Snellman $SU(n)$ expansion, the Kimura–Takamura–Yokomakura formulas, the Lagrange–Sylvester and Cayley–Hamilton polynomial forms, the adjugate method, and the eigenvector–eigenvalue identity—are projections of a single object: the resolvent $\text{adj}(\psi \mathbf{1} - \mathbf{H})/\chi'(\psi)$ evaluated at the eigenvalues. This paper makes that statement precise, gives the dictionary that translates between the methods, and extends the construction to four flavours.

For $n = 4$ we give the closed-form solution in the general-Hamiltonian style: the three $SU(4)$ invariants from traces, a characteristic quartic solved through a resolvent cubic that inherits the $SU(3)$ trigonometric formula, and a star-product tower one rung longer because the $SU(3)$ identity $(\mathbf{h} * \mathbf{h}) * \mathbf{h} = \frac{1}{3}|\mathbf{h}|^2\mathbf{h}$ is an accident of $n = 3$. A validated implementation accompanies the paper, together with the first numerical test of the n -flavour Naumov–Harrison–Scott identities.

Two results are practical. A conditioning map over the 3+1 parameter plane shows the closed form losing eight orders of magnitude in the near-degenerate band while a backward-stable eigensolver does not; the error originates in the root assembly rather than the polynomial evaluation, so the standard divided-difference remedy—which we implement and test—does not repair it. And exact gradients $\partial P/\partial\theta$ follow from implicit differentiation of the characteristic polynomial, with no eigenvector differentiated at any point.

The same construction has independent origins outside neutrino physics—in applied numerical linear algebra, magnetic resonance, multilevel quantum optics and lattice gauge theory—and we survey them for what they hold rather than for their chronology: lattice gauge theory has a published solution to the conditioning problem above and computes these derivatives routinely, magnetic resonance has a mature discipline for engineering Magnus truncations, and the closed-form $SU(n)$ exponential is known well past $n = 4$. None of that technique is currently used in neutrino physics.

Keywords: neutrino oscillations · matter effects · Cayley–Hamilton · Lagrange–Sylvester interpolation · adjugate · eigenvector–eigenvalue identity · $SU(n)$ coherence vector · Magnus expansion · sterile neutrinos

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1 Introduction

The problem is elementary to state. Given a Hermitian flavour Hamiltonian $\mathbf{H}$ of size n , constant over a baseline L , compute $P_{\nu_\alpha \rightarrow \nu_\beta} = |(e^{-i\mathbf{H}L})_{\beta\alpha}|^2$. It is the inner loop of every oscillation fit, and the field has produced a remarkable number of exact solutions to it: the Barger–Whisnant–Pakvasa–Phillips and Zaglauer–Schwarzer matter eigenvalues [1, 2], the Kimura–Takamura–Yokomakura closed formulas [3, 4], the Ohlsson–Snellman $\text{SU}(n)$ generator expansion [5, 6] implemented in `NuOscProbExact` [7], the Harrison–Scott Lagrange-form invariants [8], the four-flavour Cayley–Hamilton treatments of Kamo *et al.* [9] and Li *et al.* [10], the eigenvector–eigenvalue identity [11] and its neutrino application [12], and the adjugate method of Abdullahi and Parke [13].

These are usually presented as competing methods. They are not. Each is a projection of the single object

$$P_m = \prod_{l \neq m} \frac{\mathbf{H} - \psi_l \mathbf{1}}{\psi_m - \psi_l} = \frac{\text{adj}(\psi_m \mathbf{1} - \mathbf{H})}{\chi'(\psi_m)}, \quad e^{-i\mathbf{H}L} = \sum_{m=1}^n e^{-i\psi_m L} P_m, \quad (1)$$

with $\chi(\psi) = \det(\psi \mathbf{1} - \mathbf{H})$. Keeping the whole matrix in a generator basis gives Ohlsson–Snellman; keeping its columns gives the adjugate method; keeping its diagonal gives the eigenvector–eigenvalue identity; writing it as a polynomial with Vandermonde-determined coefficients gives the Cayley–Hamilton form. What differs is which slice is kept and in which basis, not the mathematics.

Once that is seen, two things follow. The first is that the construction extends to four flavours without new ideas, and $n = 4$ is where it stops: Abel–Ruffini removes the closed-form eigenvalues at $n \geq 5$, so the four-flavour case is the last one available in radicals. This matters because 3+1 sterile scenarios were excluded from Ref. [7] as “leaky”, when in fact they are closed and unitary over $(\nu_e, \nu_\mu, \nu_\tau, \nu_s)$ and sit squarely inside the method’s assumptions. Sections 4–6 carry this out and, more usefully, establish where it should not be trusted.

The second is that the construction has independent origins outside neutrino physics: applied numerical linear algebra (Putzer 1966; Moler and Van Loan 1978; Harris, Fillmore

and Smith 2001), magnetic resonance (Waugh, Huber and Haeberlen 1968), multilevel quantum optics (Hioe and Eberly 1981; Aravind 1986) and lattice gauge theory (Morningstar and Peardon 2004; Lüscher 2009). In the first three the neutrino version came later; lattice gauge theory is a different case, arriving at the construction at about the same time as this field and then carrying it considerably further. That ordering is worth stating once and then using, because where another field arrived first it has usually kept going: Sec. 9 collects what is sitting there.

The clearest case is internal to neutrino physics. Kim, Kim and Sze wrote the SU(3) Gell-Mann expansion for neutrinos in 1988 [?], and it is that paper the collective-oscillation literature still cites for it [?], while the exact-probability literature grew instead from Ohlsson and Snellman’s of 1999: two branches of one field, using the same expansion, tracing it to different ancestors. Within the four-flavour problem the characteristic quartic has been rediscovered three times [9, 10, 13]. This is the practical argument for a dictionary.

What this paper contains. A general- n statement of the construction and the dictionary relating its projections (Secs. 2–3); the $n = 4$ case worked out in invariant form with probability tables and a validated implementation (Secs. 4–5); a numerical study establishing where the closed form fails, why, and what does and does not repair it (Sec. 6); the surrounding territory of approximations, invariants and extensions (Sec. 7); a survey of the fields the construction also grew in and of the technique they hold (Sec. 9); and practical recommendations (Sec. 10). The algebra is not new, and where a prior source exists we cite it. What is offered is organisation, measurement, and one implementation.

2 The resolvent at its poles

2.1 Setup and conventions

Let $\mathbf{H}$ be Hermitian $n \times n$ in the flavour basis, in units making $\mathbf{H}L$ dimensionless; no decomposition into vacuum and matter parts is assumed. Expand

$$\mathbf{H} = h_0 \mathbf{1} + h_a \lambda^a, \quad h_0 = \frac{1}{n} \text{Tr } \mathbf{H}, \quad h_a = \frac{1}{2} \text{Tr}(\mathbf{H} \lambda^a), \quad a = 1, \dots, n^2 - 1, \quad (2)$$

with λ^a the generalized Gell-Mann matrices, $\text{Tr}(\lambda^a \lambda^b) = 2\delta_{ab}$ and $\lambda^a \lambda^b = \frac{2}{n} \delta_{ab} \mathbf{1} + (d_{abc} + if_{abc}) \lambda^c$. Write $\tilde{\mathbf{H}} = \mathbf{H} - h_0 \mathbf{1}$, drop the global phase $e^{-ih_0 L}$, and define the star product $(\mathbf{h} * \mathbf{h})_a \equiv d_{abc} h^b h^c$. Then

$$U(L) = e^{-i\tilde{\mathbf{H}}L} = u_0 \mathbf{1} + u_a \lambda^a, \quad P_{\nu_\alpha \rightarrow \nu_\beta} = |u_0 \delta_{\beta\alpha} + u_a (\lambda^a)_{\beta\alpha}|^2. \quad (3)$$

We take ψ_m to be the eigenvalues of $+\tilde{\mathbf{H}}$; Ref. [7] uses those of $-\tilde{\mathbf{H}}$ and factors an i from the coefficients, the two mapping onto each other by $\psi \rightarrow -\psi$, $u_k \rightarrow iu_k$. Table 1 records the notation and where it collides with the conventions of the papers we compare against.

2.2 The general- n statement

SU(n) has rank $n - 1$, so $\tilde{\mathbf{H}}$ carries $n - 1$ independent invariants, all obtainable from traces of its powers; by Newton’s identities these are the coefficients of χ (App. C).

Symbol	Here	Note / clash with the literature
$\mathbf{H}, \tilde{\mathbf{H}}$	Hamiltonian, traceless part	
ψ_m	eigenvalues of $\tilde{\mathbf{H}}$	Refs. [12, 13] write λ_i for these; we reserve λ for generators
λ^a	$\text{SU}(n)$ generators	clashes with the eigenvalue notation above; unavoidable, since both are standard
h_0, h_a	expansion coefficients of $\mathbf{H}$	
u_0, u_a	expansion coefficients of U	Ref. [7] factors an i out and uses $-\tilde{\mathbf{H}}$; map by $\psi \rightarrow -\psi, u_k \rightarrow iu_k$
I_2, I_3, I_4	$\text{SU}(n)$ invariants	$= \mathbf{h} ^2, \langle \mathbf{h}\mathbf{h}\mathbf{h} \rangle, \mathbf{h} * \mathbf{h} ^2$
$\chi(\psi)$	characteristic polynomial of $\tilde{\mathbf{H}}$	
P_m	spectral projector	$= \text{adj}(\psi_m \mathbf{1} - \tilde{\mathbf{H}}) / \chi'(\psi_m)$
$V_{\alpha i}$	mixing matrix in matter	$U_{\alpha i}$ in vacuum
L	baseline	phases are $\psi_m L$, dimensionless

Table 1: Notation, and where it collides with the conventions of the papers we compare against.

The construction has four steps, and each is the point at which one of the methods of Sec. 3 branches off. (i) For a non-degenerate Hermitian $\tilde{\mathbf{H}}$ the spectral decomposition is $U = \sum_m e^{-i\psi_m L} P_m$ with P_m the orthogonal projector onto the m -th eigenspace. (ii) The polynomial $\prod_{l \neq m} (\tilde{\mathbf{H}} - \psi_l \mathbf{1})$ annihilates every eigenvector except the m -th, on which it acts as $\prod_{l \neq m} (\psi_m - \psi_l) = \chi'(\psi_m)$; hence $P_m = \prod_{l \neq m} (\tilde{\mathbf{H}} - \psi_l \mathbf{1}) / \chi'(\psi_m)$, which is Sylvester's formula. (iii) The same object is the adjugate: away from the spectrum, $\text{adj}(\psi \mathbf{1} - \tilde{\mathbf{H}}) = \chi(\psi)(\psi \mathbf{1} - \tilde{\mathbf{H}})^{-1} = \prod_k (\psi - \psi_k) \sum_l P_l / (\psi - \psi_l) = \sum_l P_l \prod_{k \neq l} (\psi - \psi_k)$, a polynomial identity, and setting $\psi = \psi_m$ kills every term but $l = m$:

$$\text{adj}(\psi_m \mathbf{1} - \tilde{\mathbf{H}}) = \chi'(\psi_m) P_m. \quad (4)$$

(iv) Contracting with the trace metric, $u_0 = \frac{1}{n} \text{Tr} U$ and $u_a = \frac{1}{2} \text{Tr}(\lambda^a U)$, and expanding the adjugate in powers of $\tilde{\mathbf{H}}$ by Faddeev–LeVerrier, $\text{adj}(\psi \mathbf{1} - \tilde{\mathbf{H}}) = \sum_{j=0}^{n-1} \psi^{n-1-j} B_j$ with $B_0 = \mathbf{1}$ and $B_j = \tilde{\mathbf{H}} B_{j-1} + d_j \mathbf{1}$, converts each power into a star-tower vector via App. B. The result is

$$u_0 = \frac{1}{n} \sum_m e^{-i\psi_m L}, \quad u_a = \sum_m e^{-i\psi_m L} \frac{N_a(\psi_m)}{\chi'(\psi_m)}, \quad N_a(\psi) \equiv \frac{1}{2} \text{Tr}[\lambda^a \text{adj}(\psi \mathbf{1} - \tilde{\mathbf{H}})], \quad (5)$$

The structure of N_a follows without case analysis. Faddeev–LeVerrier gives $\text{adj}(\psi \mathbf{1} - \tilde{\mathbf{H}}) = \sum_{j=0}^{n-1} \psi^{n-1-j} B_j$ with $B_0 = \mathbf{1}$, $B_j = \tilde{\mathbf{H}} B_{j-1} + d_j \mathbf{1}$ and $d_j = -\frac{1}{j} \text{Tr}(\tilde{\mathbf{H}} B_{j-1})$, so each B_j is a polynomial in $\tilde{\mathbf{H}}$ of degree j with invariant coefficients. Projecting, $\frac{1}{2} \text{Tr}(\lambda^a B_0) = 0$ and $\frac{1}{2} \text{Tr}(\lambda^a B_j)$ for $j \geq 1$ is a combination of $\frac{1}{2} \text{Tr}(\lambda^a \tilde{\mathbf{H}}^k)$, $k \leq j$, which by App. B is the k -th star-tower vector dressed by invariants. Hence $N_a(\psi)$ has degree $n-2$ in ψ —the ψ^{n-1} coefficient vanishes because $\text{Tr} \lambda^a = 0$ —and its coefficients are $\mathbf{h}, \mathbf{h} * \mathbf{h}, (\mathbf{h} * \mathbf{h}) * \mathbf{h}, \dots$, a tower of length $n-1$, the last rung being $\frac{1}{2} \text{Tr}(\lambda^a \tilde{\mathbf{H}}^{n-1})$ minus invariant multiples of lower rungs. Cayley–Hamilton eliminates $\tilde{\mathbf{H}}^n$ and above, which is why the tower stops. This is the

general statement of which Eqs. (11), (12) and (20) are the cases $n = 2, 3, 4$; App. E lists them.

Everything follows from traces of powers of $\tilde{\mathbf{H}}$: no d -tensor at runtime, no principal minors, no parametrization, and—the structural point emphasised in Refs. [12, 14]—no eigenvectors. The eigenvalues are the only non-polynomial input.

One caveat should be stated plainly, since “no eigenvectors needed” is easy to oversell. Probabilities need none. But the mixing matrix *in matter*, which is what one wants for oscillograms, resonance conditions, matter-dressed angles and the invariants of Sec. 7.2, is eigenvector information, and obtaining it means going to projection (b)—the adjugate columns—or to projection (c) for the moduli alone. Route (a) buys economy by discarding exactly the structure that makes matter effects interpretable.

2.3 Degeneracy

Equation (5) assumes distinct ψ_m . Exact degeneracy is measure-zero for random $\mathbf{H}$, but that is a misleading way to describe the situation: level crossings are precisely where resonances happen, so the near-degenerate region is where oscillation physics is most interesting and where every evaluation route is worst conditioned. The correct generalization is the confluent (Hermite) interpolation: if ψ is a root of χ of multiplicity μ , one matches f and its first $\mu - 1$ derivatives there, and

$$U = \sum_{\text{distinct } \psi} \sum_{k=0}^{\mu_\psi-1} \frac{(-iL)^k e^{-i\psi L}}{k!} (\tilde{\mathbf{H}} - \psi \mathbf{1})^k \Pi_\psi, \quad (6)$$

with Π_ψ the projector onto the full eigenspace. For Hermitian $\tilde{\mathbf{H}}$ the eigenspaces are orthogonal and the sum over k truncates at $k = 0$, so mathematically the degenerate limit is benign: $U = \sum_{\text{distinct}} e^{-i\psi L} \Pi_\psi$.

Numerically it is not benign at all unless the implementation is told about it. Evaluated as written, Eq. (1) divides by $\psi_m - \psi_l = 0$ and returns `inf`. The remedy is a merge tolerance: cluster roots separated by less than τ times the spectral spread, replace each cluster by its mean, and take the Lagrange product over distinct clusters only, which is Eq. (6) with μ_ψ read off from the clustering. With $\tau = 10^{-9}$ this recovers machine precision on constructed spectra with two-, three- and four-fold degeneracies— $\{1, 1, -\frac{1}{2}, -\frac{3}{2}\}$, $\{2, 2, 2, -6\}$, $\{1, 1, -1, -1\}$ and $\{3, -1, -1, -1\}$ —where the unguarded form fails outright. Any implementation of Eq. (5) needs this test; ours did not have it until this study.

What remains is the *near*-degenerate case, where $\chi'(\psi_m)$ is small but non-zero, no clustering is legitimate, and the formula is evaluated as a difference of large terms. Section 6 quantifies it.

2.4 Equivalent evaluations

Four evaluations of Eq. (1) recur, with different cost and different numerical character. They are Methods 8, 10 and 14 of the Moler–Van Loan taxonomy [15] in that paper’s language.

- (i) *Lagrange product*: $\prod_{l \neq m} (\tilde{\mathbf{H}} - \psi_l \mathbf{1})$, requiring $n - 1$ matrix multiplications per mode.

- (ii) *Adjugate / Faddeev–LeVerrier*: generate the coefficient matrices of $\text{adj}(\psi\mathbf{1} - \tilde{\mathbf{H}})$ once, then evaluate at each ψ_m [16, 17]. Abdullahi and Parke [13] prove the identity of (i) and (ii) and note that $\chi'(\psi_m) = \text{Tr} \text{adj}(\psi_m\mathbf{1} - \mathbf{H})$, so a single eigenvalue suffices to build a single projector.
- (iii) *Cayley–Hamilton polynomial*: $U = \sum_{j=0}^{n-1} c_j \tilde{\mathbf{H}}^j$ with $\sum_j c_j \psi_m^j = e^{-i\psi_m L}$, an $n \times n$ Vandermonde solve. Fastest in our tests, and the most ill-conditioned in principle [15].
- (iv) *Putzer recursion* [18]: eigenvalues-only, equivalent to (iii) but organized to avoid the explicit Vandermonde inverse.

3 One object, four projections and two extensions

The six items of Fig. 1 are not six things of the same kind, and the distinction is worth stating precisely.

Items (a)–(d) are projections in the strict sense: the same matrix P_m , or the same operator U , written in a different basis or restricted to a sub-block. Item (e) is not a projection but the *induced adjoint action*: the map $U \mapsto U \otimes U^*$ carrying the fundamental representation to the adjoint, under which the evolution of the density matrix is the image of the evolution of the state. It is a functor, not a restriction, and it is invertible only up to a global phase. Item (f) is neither: the Magnus expansion solves a *different problem*, time-ordered evolution under non-commuting $\mathbf{H}(x)$, of which the constant-density construction is an ingredient—the exponential map applied to each truncation—rather than a special case. The reduction runs the other way: constant $\mathbf{H}$ is the degenerate instance in which the Magnus series terminates at first order.

(a) $\text{SU}(n)$ generator expansion = the whole matrix, in the generator basis. Contracting Eq. (1) with the trace metric gives Eq. (5) directly:

$$u_0 = \frac{1}{n} \text{Tr} U, \quad u_a = \frac{1}{2} \text{Tr}(\lambda^a U) = \sum_m e^{-i\psi_m L} \frac{\frac{1}{2} \text{Tr}[\lambda^a \text{adj}(\psi_m \mathbf{1} - \tilde{\mathbf{H}})]}{\chi'(\psi_m)}. \quad (7)$$

The Ohlsson–Snellman coefficients are therefore the $\text{SU}(n)$ -basis components of the adjugate: the two methods differ by a change of basis and nothing else. We record this because the relationship has been described as distant [13, Sec. III.A]; the distance is one trace. The same expansion, with the same invariants, is the Hioe–Eberly coherence vector [19], and the same Cayley–Hamilton solution of it is Aravind’s [20].

(b) Adjugate method = the columns. $V_{\alpha i} V_{\beta i}^* = (P_i)_{\alpha\beta}$, read off in the flavour basis. This is the natural resolution for matter mixing elements and matter-potential invariants, and the one at which the n -flavour generalization of the Naumov–Harrison–Scott identity [21, 22] was derived [13]. The Kimura–Takamura–Yokomakura construction [3, 4] is projection (b) with the ψ -dependence resummed: they solve the linear system relating $|V_{\alpha i}|^2$ to $\{1, \lambda_i, \lambda_i^2\}$ obtained from the diagonal of adj , which is Eq. (8) rearranged, and then reconstruct the off-diagonal bilinears from unitarity plus one invariant. It therefore keeps the columns like (b) but obtains them from the diagonal like (c), which is why it is often described as intermediate.

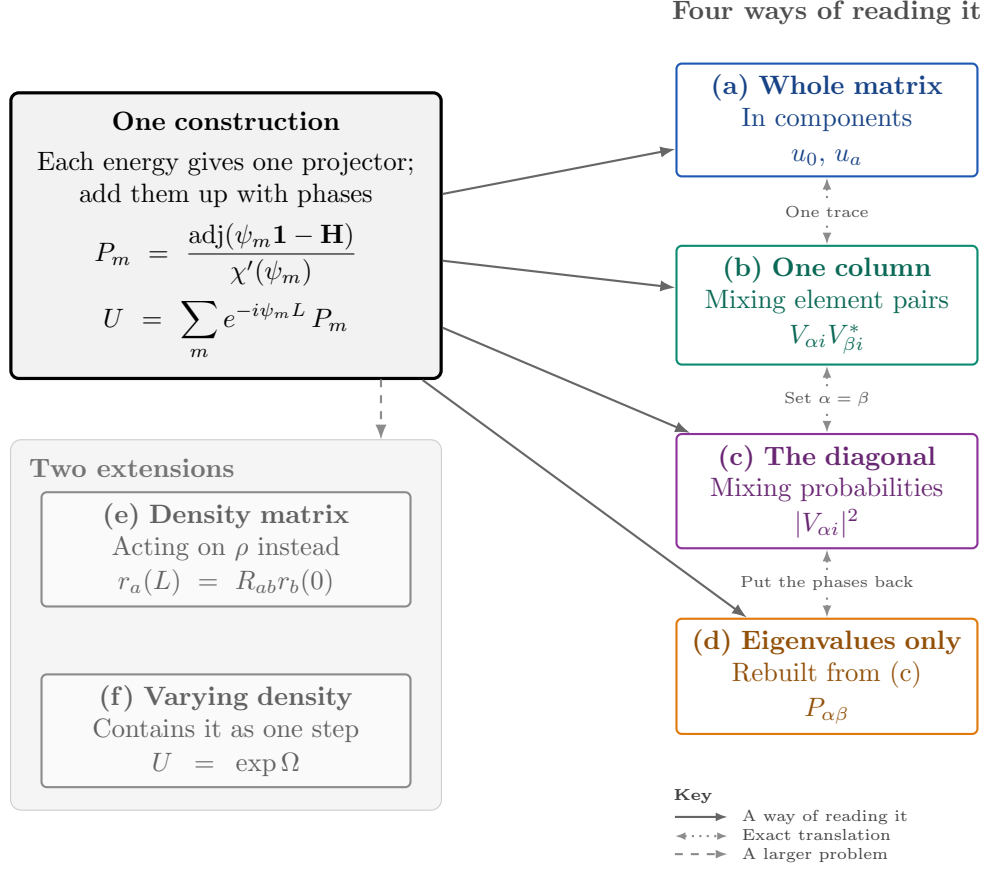


Figure 1: **Structure.** Items (a)–(d) are four readings of a single object: the same operator written in a different basis, or restricted to part of itself. They are not competing methods, and choosing among them changes nothing that can be measured. Item (e) is not a reading but the same operator acting on density matrices; item (f) solves the harder problem of a Hamiltonian that changes along the path, and reduces to the construction above when it does not. Where each of these came from is deliberately left out here; see Sec. 9.

(c) Eigenvector–eigenvalue identity = the diagonal. The α -th diagonal entry of $\text{adj}(\psi \mathbf{1} - \mathbf{H})$ is the characteristic polynomial of the principal minor M_α , so

$$|V_{\alpha m}|^2 \prod_{l \neq m} (\psi_m - \psi_l) = \prod_k (\psi_m - \mu_k^{(\alpha)}), \quad (8)$$

which is the identity of Ref. [11]. Restricting to the diagonal discards all relative phase information; the equivalence is stated in [13].

(d) Eigenvalue-only probabilities = diagonal + phase reconstruction. Reference [12] recovers matter mixing angles from Eq. (8) and restores the discarded phases through supplementary identities—Toshev [23] for δ in matter, Naumov–Harrison–Scott for the Jarlskog factor—then reassembles probabilities, exactly or with DMP-quality approximate eigenvalues [24, 25]. It rebuilds, through the standard parametrization, what route (a) never took apart.

(e) **Adjoint representation = the density-matrix extension.** SQuIDS [26] expands $\rho = r_0 \mathbf{1} + r_a \lambda^a$ and integrates $\dot{r}_a = f_{abc} h_b r_c$ plus non-unitary terms; nuSQuIDS [27] builds neutrino propagation on this up to $n = 6$. For a closed system at constant density the exact flow map is

$$r_a(L) = R_{ab}(L) r_b(0), \quad R_{ab} = \frac{1}{2} \text{Tr} (\lambda^a U \lambda^b U^\dagger) \in SO(n^2 - 1), \quad (9)$$

built entirely from the (u_0, u_a) of Eq. (5). The Ohlsson–Snellman method is thus the analytic kernel of the SQuIDS equations, restricted to closed systems and constant density; numerical integration buys time dependence, interactions and decoherence in exchange. The precession equation (9) is Hioe and Eberly’s rotating coherence vector [19], and the extension to relaxation is Aravind’s [20].

(f) **Magnus expansion = the x -dependent extension.** For a profile $\mathbf{H}(x)$, Magnus [28] writes $U(L) = \exp \Omega(L)$ with $\Omega = \sum_k \Omega_k$ and [29]

$$\Omega_1 = -i \int_0^L \mathbf{H}(x) dx, \quad \Omega_2 = -\frac{1}{2} \int_0^L dx \int_0^x dy [\mathbf{H}(x), \mathbf{H}(y)], \quad \dots \quad (10)$$

For constant $\mathbf{H}$ the series terminates at Ω_1 . And every truncation $\Omega^{[k]}$ is anti-Hermitian, hence (after removing the trace) an element of $\mathfrak{su}(n)$, because nested commutators never leave the algebra; therefore $i\Omega^{[k]}/L = h_a^{\text{eff}} \lambda^a$ and $\exp \Omega^{[k]}$ is again evaluable exactly by Eq. (5) with $h_a \rightarrow h_a^{\text{eff}}$. This is why Magnus truncations are unitary at every order, the property emphasised for neutrinos by Ioannian and Smirnov [30] and present since the first applications to non-adiabatic crossing [31]. In the language of geometric integration [32], Magnus supplies the algebra element and Eq. (5) supplies the exponential map; the special-purpose integrator of Casas *et al.* [33] is such a scheme, and the multi-slab product of Ref. [7, Sec. VIB] is its lowest-order, piecewise-constant member.

The classical layer. The six items above are modern, but the object they read is not, and its antecedents do not divide six ways either. Sylvester’s interpolation formula [34] is Eq. (1); Frobenius’s covariants (1878) are the projectors P_m themselves; the identity behind (c) is a corollary, due to Jacobi (1834), of the same statement about the adjugate that (b) uses; Cayley–Hamilton (1858) supplies the polynomial closure that (d) inverts; and the fundamental and adjoint representations that separate (a) from (e) are one construction in Lie’s sense. The classical material is therefore a single connected cluster, essentially complete by the 1890s, from which all six readings descend jointly rather than along six separate lines. That is the same statement as the one made above for the modern methods, taken one layer down, and it is why the unification is available at all: the nineteenth century had already done it, and what the twentieth and twenty-first centuries added was the physics each reading is convenient for.

4 Worked cases

4.1 $n = 2$

With $\lambda^a = \sigma^a$ there is one invariant $|\mathbf{h}|$, the star product vanishes, $\psi_{\pm} = \pm|\mathbf{h}|$, and Eq. (5) collapses to

$$U = \cos(|\mathbf{h}|L) \mathbf{1} - i \sin(|\mathbf{h}|L) \frac{h_a}{|\mathbf{h}|} \sigma^a, \quad (11)$$

i.e. Eqs. (4)–(6) of Ref. [7]. The dictionary is visible by inspection: $\text{adj}(\psi \mathbf{1} - \tilde{\mathbf{H}})_{\alpha\alpha} = \psi - \tilde{\mathbf{H}}_{\beta\beta}$ is the characteristic polynomial of the 1×1 minor, so Eq. (8) reads $|V_{\alpha\pm}|^2 = \frac{1}{2}(1 \pm \tilde{\mathbf{H}}_{\alpha\alpha}/|\mathbf{h}|)$ —the two-flavour mixing formula, with no mixing angle ever written.

4.2 $n = 3$

The eigenvalues solve $\psi^3 - |\mathbf{h}|^2\psi - \frac{2}{3}\langle\mathbf{h}\mathbf{h}\mathbf{h}\rangle = 0$ by the trigonometric formula, and

$$u_0 = \frac{1}{3} \sum_m e^{-i\psi_m L}, \quad u_a = \sum_m e^{-i\psi_m L} \frac{\psi_m h_a + (\mathbf{h} * \mathbf{h})_a}{3\psi_m^2 - |\mathbf{h}|^2}, \quad (12)$$

with denominator $\chi'(\psi_m)$. Two accidents make $n = 3$ compact: rank 2, so only $|\mathbf{h}|^2$ and $\langle\mathbf{h}\mathbf{h}\mathbf{h}\rangle = d_{abc}h^a h^b h^c$ appear; and

$$(\mathbf{h} * \mathbf{h}) * \mathbf{h} = \frac{1}{3}|\mathbf{h}|^2 \mathbf{h} \quad (\text{SU}(3) \text{ only}), \quad (13)$$

truncating the star tower at one rung so that $N_a(\psi) = \psi h_a + (\mathbf{h} * \mathbf{h})_a$ exactly. Both fail at $n = 4$.

4.3 $n = 4$: the last closed form

SU(4) has rank 3; the invariants are

$$I_2 = |\mathbf{h}|^2 = \frac{1}{2} \text{Tr} \tilde{\mathbf{H}}^2, \quad I_3 = \langle\mathbf{h}\mathbf{h}\mathbf{h}\rangle = \frac{1}{2} \text{Tr} \tilde{\mathbf{H}}^3, \quad I_4 = |\mathbf{h} * \mathbf{h}|^2 = \frac{1}{2} (\text{Tr} \tilde{\mathbf{H}}^4 - I_2^2), \quad (14)$$

and the star tower follows by projecting the general identities of App. B,

$$\tilde{\mathbf{H}}^2 = \frac{2}{n} I_2 \mathbf{1} + (\mathbf{h} * \mathbf{h})_a \lambda^a, \quad \tilde{\mathbf{H}}^3 = \frac{2}{n} I_3 \mathbf{1} + \left[\frac{2}{n} I_2 h_a + ((\mathbf{h} * \mathbf{h}) * \mathbf{h})_a \right] \lambda^a, \quad (15)$$

so that at $n = 4$

$$(\mathbf{h} * \mathbf{h})_a = \frac{1}{2} \text{Tr}(\lambda^a \tilde{\mathbf{H}}^2), \quad ((\mathbf{h} * \mathbf{h}) * \mathbf{h})_a = \frac{1}{2} \text{Tr}(\lambda^a \tilde{\mathbf{H}}^3) - \frac{1}{2} I_2 h_a. \quad (16)$$

The second rung is independent data, and App. B shows why: at $n = 3$ the Cayley–Hamilton relation $\tilde{\mathbf{H}}^3 = I_2 \tilde{\mathbf{H}} + \frac{2}{3} I_3 \mathbf{1}$ combined with Eq. (15) forces Eq. (13), whereas at $n = 4$ Cayley–Hamilton constrains $\tilde{\mathbf{H}}^4$, hence $(\mathbf{h} * \mathbf{h}) * (\mathbf{h} * \mathbf{h})$, and leaves $(\mathbf{h} * \mathbf{h}) * \mathbf{h}$ free. Numerically Eq. (13) is violated at the $\mathcal{O}(40\%)$ level for random $\mathbf{h}$.

The characteristic polynomial is the depressed quartic

$$\chi(\psi) = \psi^4 - I_2 \psi^2 - \frac{2}{3} I_3 \psi + \frac{1}{4} (I_2^2 - 2I_4) = 0, \quad (17)$$

solved (Ferrari–Euler) through the resolvent cubic

$$z^3 - 2I_2z^2 + 2I_4z - \frac{4}{9}I_3^2 = 0, \quad (18)$$

whose roots $z_i = (\psi_j + \psi_k)^2$ are real and non-negative because $\tilde{\mathbf{H}}$ is Hermitian—so Eq. (18) yields to *exactly* the trigonometric cosine formula of the $n = 3$ case, and the three-flavour machinery sits inside the four-flavour solution. Then

$$\psi_m = \frac{1}{2}(s_1\sqrt{z_1} + s_2\sqrt{z_2} + s_3\sqrt{z_3}), \quad s_i = \pm 1, \quad s_1s_2s_3\sqrt{z_1z_2z_3} = \frac{2}{3}I_3, \quad (19)$$

This closed form is not new outside neutrino physics. Kaiser [?] solves the SU(4) matrix exponential by exactly this route—quartic, resolvent cubic with three positive roots, irreducible case, trisection of an angle—and sketches SU(5), SU(6) and Sp(3); Van Kortryk [35] gives the SU(4) and SU(5) exponentials explicitly in terms of $\text{Tr } \tilde{\mathbf{H}}^2$, $\text{Tr } \tilde{\mathbf{H}}^3$ and $\text{Tr } \tilde{\mathbf{H}}^4$, i.e. in the invariants of Eq. (14). The division of labour is therefore clean: the closed-form exponential is theirs, and what follows here—the probability tables of Sec. 5, the conditioning analysis and its diagnosis, the degeneracy handling, the density gradient, and a general- $\mathbf{H}_4$ implementation—is not in any of those references, nor is the observation that none of them is cited by the neutrino literature that needs them.

With that acknowledged, Eq. (5) becomes

$$u_0 = \frac{1}{4} \sum_{m=1}^4 e^{-i\psi_m L}, \quad u_a = \sum_{m=1}^4 e^{-i\psi_m L} \frac{(\psi_m^2 - \frac{1}{2}I_2)h_a + \psi_m(\mathbf{h} * \mathbf{h})_a + ((\mathbf{h} * \mathbf{h}) * \mathbf{h})_a}{4\psi_m^3 - 2I_2\psi_m - \frac{2}{3}I_3}. \quad (20)$$

Physics scope. Over the full basis $(\nu_e, \nu_\mu, \nu_\tau, \nu_s)$ a 3+1 system is closed and unitary. In constant matter, after removing the flavour-universal neutral-current potential as a global phase, the potential matrix is [9, 10, 13]

$$\mathbf{A}_4 = \text{diag}(V_{CC}, 0, 0, V_s), \quad V_{CC} = \sqrt{2}G_F n_e, \quad V_s = -V_{NC} = +\frac{G_F n_n}{\sqrt{2}}, \quad (21)$$

so $V_s \simeq V_{CC}/2$ in the crust. Because Eq. (20) takes an arbitrary Hermitian $\mathbf{H}_4$, it covers short-baseline 3+1 fits, the TeV-scale Earth resonance of eV-scale steriles, 3+1+NSI and 3+1+LIV with no further derivation. Figure 2 shows the two regimes that motivate the extension.

5 Probability tables

The probabilities follow from Eq. (3) once the generator basis is fixed, and they take the same shape at every n . For each unordered flavour pair (α, β) , $\alpha < \beta$, exactly two generators are supported on it: a symmetric one, with unit entries in positions (α, β) and (β, α) , and an antisymmetric one, with $-i$ and $+i$. Denote their coefficients $u_{\alpha\beta}^S$ and $u_{\alpha\beta}^A$; these are simply two of the u_a , relabelled by the pair they act on. The remaining $n - 1$ generators are

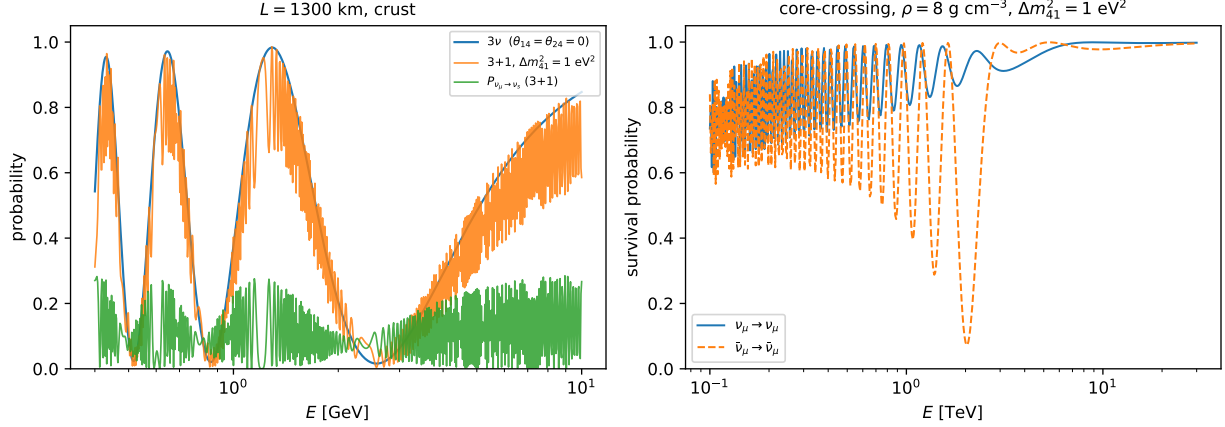


Figure 2: **3+1 in scope**. Left: at a long-baseline configuration, the sterile state superimposes a rapid oscillation on $P_{\nu_\mu \rightarrow \nu_\mu}$ and opens the $\nu_\mu \rightarrow \nu_s$ channel that the three-flavour formalism cannot represent. Right: for a core-crossing chord the sterile matter potential drives a resonance at TeV energies, appearing in the antineutrino channel for $\Delta m_{41}^2 > 0$ and reaching $P_{\bar{\nu}_\mu \rightarrow \bar{\nu}_\mu} = 0.07$ near 2 TeV, with no counterpart for neutrinos—the signature searched for in atmospheric data. Both panels use Eq. (20) throughout; the resonance energy $E \simeq \Delta m_{41}^2 / 2V_{\text{NC}}$ places it where the standard estimate does.

the Cartan matrices $D_l = \sqrt{2/[l(l+1)]} \text{diag}(1, \dots, 1, -l, 0, \dots, 0)$ with l leading ones, and coefficients $u^{(l)}$. Then $U_{\beta\alpha} = u_{\alpha\beta}^S + iu_{\alpha\beta}^A$ for $\alpha < \beta$, and

$$\boxed{\begin{aligned} P_{\nu_\alpha \rightarrow \nu_\beta} &= |u_{\alpha\beta}^S + iu_{\alpha\beta}^A|^2, & P_{\nu_\beta \rightarrow \nu_\alpha} &= |u_{\alpha\beta}^S - iu_{\alpha\beta}^A|^2 & (\alpha < \beta), \\ P_{\nu_\alpha \rightarrow \nu_\alpha} &= \left| u_0 + \sum_{l=1}^{n-1} (D_l)_{\alpha\alpha} u^{(l)} \right|^2. \end{aligned}} \quad (22)$$

Three structural statements follow at once, for any n . The $n(n-1)$ off-diagonal entries come in T -conjugate pairs differing only by the sign of the antisymmetric coefficient, so $P_{\nu_\alpha \rightarrow \nu_\beta} - P_{\nu_\beta \rightarrow \nu_\alpha} = 4\Im(u_{\alpha\beta}^S \overline{u_{\alpha\beta}^A})$ and T violation resides entirely there; for $n=2$ the star product vanishes, u^S and u^A share a phase, and the difference is identically zero. Unitarity is not imposed but inherited, appearing as the n constraints $\sum_\beta P_{\nu_\alpha \rightarrow \nu_\beta} = 1$, i.e. $|u_0|^2 + \sum_a |u_a|^2 = 1/n$ together with $\Re(\bar{u}_0 u_a) + \frac{1}{2} d_{abc} \bar{u}_b u_c = 0$ —useful as a runtime check. And the number of physical Dirac phases entering the off-diagonal entries is $(n-1)(n-2)/2$: none at $n=2$, one at $n=3$, three at $n=4$.

Antineutrinos. These follow without new work. $\mathbf{H} \rightarrow \mathbf{H}^*$ with $V \rightarrow -V$ sends $h_a \rightarrow h_a$ for the symmetric generators and $h_a \rightarrow -h_a$ for the antisymmetric ones, hence $u_{\alpha\beta}^A \rightarrow -u_{\alpha\beta}^A$ with u^S and the Cartan coefficients unchanged. Table 2 therefore gives the antineutrino probabilities by exchanging the two members of each T -conjugate pair, and CP violation appears as the same $\Im(u^S \bar{u}^A)$ that carries T violation. All numerical statements in Sec. 6 were checked in both channels.

For $n = 2$ and $n = 3$ the numbering is fixed by convention (Pauli, Gell-Mann). For $n = 4$ none exists, and since a published table tends to become the convention we state our choice and our reason. We order the six flavour pairs lexicographically, symmetric before antisymmetric, and place the three Cartan generators last. This reduces to the standard Gell-Mann ordering on the upper-left 3×3 block if one deletes the pairs involving the fourth state, so three-flavour formulas embed unchanged; it keeps the two generators of a given pair adjacent, so the T -conjugate structure reads off consecutive indices; and it places the sterile-singling generator λ^{15} last, so that adding a fifth state appends rather than rennumbers.

$n = 2$ (Pauli basis, $a = 1, 2, 3$)			
$\alpha \backslash \beta$	ν_e	ν_μ	
ν_e	$ u_0 + u_3 ^2$	$ u_1 + iu_2 ^2$	
ν_μ	$ u_1 - iu_2 ^2$	$ u_0 - u_3 ^2$	

$n = 3$ (Gell-Mann basis, $a = 1, \dots, 8$)			
$\alpha \backslash \beta$	ν_e	ν_μ	ν_τ
ν_e	$ u_0 + u_3 + \frac{u_8}{\sqrt{3}} ^2$	$ u_1 + iu_2 ^2$	$ u_4 + iu_5 ^2$
ν_μ	$ u_1 - iu_2 ^2$	$ u_0 - u_3 + \frac{u_8}{\sqrt{3}} ^2$	$ u_6 + iu_7 ^2$
ν_τ	$ u_4 - iu_5 ^2$	$ u_6 - iu_7 ^2$	$ u_0 - \frac{2u_8}{\sqrt{3}} ^2$

$n = 4$ (basis of App. A, $a = 1, \dots, 15$)				
$\alpha \backslash \beta$	ν_e	ν_μ	ν_τ	ν_s
ν_e	$ u_0 + u_{13} + \frac{u_{14}}{\sqrt{3}} + \frac{u_{15}}{\sqrt{6}} ^2$	$ u_1 + iu_2 ^2$	$ u_3 + iu_4 ^2$	$ u_5 + iu_6 ^2$
ν_μ	$ u_1 - iu_2 ^2$	$ u_0 - u_{13} + \frac{u_{14}}{\sqrt{3}} + \frac{u_{15}}{\sqrt{6}} ^2$	$ u_7 + iu_8 ^2$	$ u_9 + iu_{10} ^2$
ν_τ	$ u_3 - iu_4 ^2$	$ u_7 - iu_8 ^2$	$ u_0 - \frac{2u_{14}}{\sqrt{3}} + \frac{u_{15}}{\sqrt{6}} ^2$	$ u_{11} + iu_{12} ^2$
ν_s	$ u_5 - iu_6 ^2$	$ u_9 - iu_{10} ^2$	$ u_{11} - iu_{12} ^2$	$ u_0 - \frac{3u_{15}}{\sqrt{6}} ^2$

Table 2: All oscillation probabilities $P_{\nu_\alpha \rightarrow \nu_\beta}$, with the initial flavour α labelling rows and the final flavour β columns. **The coefficients u_0, u_a are those of Eq. (3) for the relevant n and are different objects in each block:** there are 3, 8 and 15 of them at $n = 2, 3, 4$, obtained from $u_a = \frac{1}{2} \text{Tr}(\lambda^a U)$ with the generators of that n , so u_3 at $n = 2$ (a Pauli coefficient) has nothing to do with u_3 at $n = 4$ (the symmetric $\nu_e \nu_\tau$ coefficient). Index labelling follows the standard Pauli and Gell-Mann conventions at $n = 2, 3$; at $n = 4$ it follows App. A, in which the six flavour pairs appear lexicographically, symmetric before antisymmetric, and the three Cartan generators last. Reverse channels are the transposed entries, differing only in the sign of the antisymmetric coefficient; antineutrino probabilities are obtained by the same transposition. The $n = 3$ block is Table IV of Ref. [7] in the convention of Sec. 2; the $n = 4$ block is its four-flavour analogue. Every entry has been verified against direct diagonalization at the 10^{-16} level.

Two features are worth noting. The sterile survival probability $P_{\nu_s \rightarrow \nu_s} = |u_0 - 3u_{15}/\sqrt{6}|^2$ involves only the identity and the last Cartan coefficient, because ν_s is the state singled

out by λ^{15} , the last Cartan generator; the same structure makes $P_{\nu_\tau \rightarrow \nu_\tau}$ the simplest of the three-flavour diagonal entries. And the twelve off-diagonal entries at $n = 4$ carry all three Dirac phases, consistent with the counting $(n-1)(n-2)/2 = 3$ of Sec. 7.2.

6 Numerical behaviour

6.1 Validation

The accompanying implementation validates both routes against `numpy.linalg.eigh` and, independently, against an arbitrary-precision reference built from a 60-digit `mpmath` Hermitian eigendecomposition. Over 2000 random Hermitian $(\mathbf{H}, L)$ the eigenvalues, the two routes, the probabilities and unitarity all agree to machine precision.

A stronger and independent test is available at $n = 4$: the generalized Naumov–Harrison–Scott relations of Sec. 7.2. These equate a vacuum quantity to a matter quantity, so they probe eigenvalues and eigenvector bilinears simultaneously and are insensitive to the evaluation route. Because a check at a single point is weak—a near-decoupled sterile state makes the identity nearly trivial—we swept 400 random configurations with $\theta_{14}, \theta_{24}, \theta_{34}$ uniform on $[0, \pi/2]$, δ uniform on $[0, 2\pi]$, Δm_{41}^2 log-uniform on $[10^{-3}, 10]$ eV² and E log-uniform on $[0.2, 50]$ GeV, evaluating all three independent channels. The identities hold to machine precision away from degeneracies, and degrade only in the near-degenerate corner—at near-maximal θ_{24}, θ_{34} with a small splitting—as Eq. (24) requires. We are not aware of a previous numerical check of these relations.

6.2 Conditioning

That the closed form is less stable near degeneracies is folklore. What follows makes it quantitative in a form that does not depend on the implementation: the cost of passing through χ is a power of the relative spectral gap, and the exponent is fixed by how many eigenvalues are close, not by the code, the library or the machine.

Throughout, $\Delta = \psi_{\max} - \psi_{\min}$ is the spectral spread, u the unit roundoff, and

$$g = \frac{\min_{i \neq j} |\psi_i - \psi_j|}{\Delta} \quad (23)$$

the relative gap. We say a *cluster of size r* is present when r eigenvalues lie within $\delta = g\Delta$ of one another. Only $2 \leq r \leq n-1$ is meaningful: if all n eigenvalues cluster then δ and Δ coincide and $g = 1$ by construction.

What forming χ costs. The coefficients of $\chi(\psi) = \psi^n + c_{n-1}\psi^{n-1} + \dots + c_0$ are formed from traces of powers of $\tilde{\mathbf{H}}$, so $c_k = O(\Delta^{n-k})$ is obtained with absolute error $O(u\Delta^{n-k})$ and the induced perturbation of χ near the spectrum is $O(u\Delta^n)$. Write $\chi = \prod_{j \in C} (\psi - \psi_j) \cdot Q(\psi)$ with C the cluster and $Q = O(\Delta^{n-r})$ away from it. A perturbation ε displaces a root of the cluster by d with $d\delta^{r-1} \sim \varepsilon/Q$ while $d \ll \delta$, and by $d^r \sim \varepsilon/Q$ once $d \gg \delta$. Hence

$$|\delta\psi| = \begin{cases} O(u\Delta g^{-(r-1)}), & g \gg u^{1/r}, \\ O(\Delta u^{1/r}), & g \lesssim u^{1/r}, \end{cases} \quad (24)$$

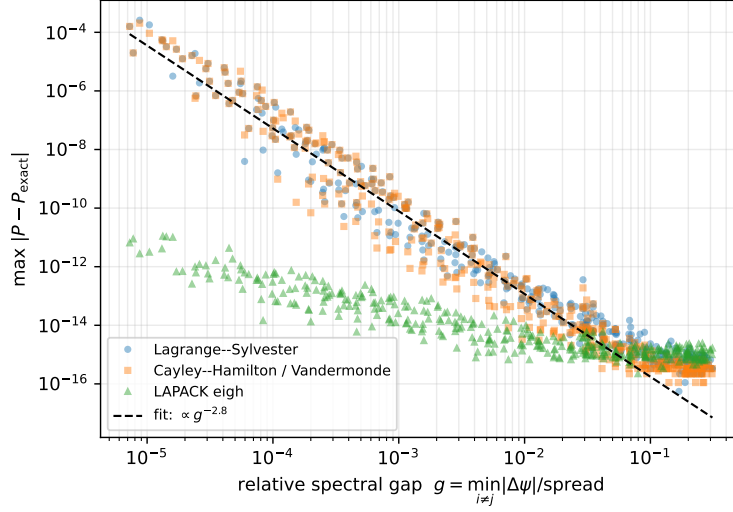


Figure 3: **The exponent is the cluster size.** Eigenvalue error, normalised by $u\Delta$ so that no machine-dependent scale remains, against the relative gap g , for spectra constructed with a prescribed cluster size r . Roots of χ follow $g^{-(r-1)}$ and flatten at the saturation gaps $u^{1/r}$; a backward-stable eigensolver is flat throughout. Fitted slopes -0.96 , -0.98 and -2.06 against the predicted -1 , -1 and -2 .

the second line being the saturation at which the cluster can no longer be resolved at all. A backward-stable eigensolver applied to $\tilde{\mathbf{H}}$ instead of to χ obeys $|\delta\psi| = O(u\Delta)$ by Weyl's inequality, for every r and every g . The penalty for passing through the characteristic polynomial is therefore the factor $g^{-(r-1)}$, and nothing else.

The exponent by flavour count. Since $r \leq n - 1$, the largest attainable exponent at n flavours is $n - 2$:

- $n = 2$: no penalty. With two eigenvalues the gap *is* the spread, $g \equiv 1$, and the two-flavour closed form is unconditionally well conditioned. Its reputation for robustness is a theorem, not an accident of use.
- $n = 3$: at worst one power, $|\delta\psi| = O(u\Delta/g)$, saturating near $g \sim u^{1/2}$.
- $n = 4$: at worst two powers, $O(u\Delta/g^2)$, saturating near $g \sim u^{1/3}$.

The exponent depends on the cluster size alone, so the same statement covers $n \geq 5$ without further work: the worst case degrades by one power per added flavour. Figure 3 verifies Eq. (24) on synthetic spectra in which r and g are set independently, which is necessary because on a physical parameter scan the cluster structure changes along the scan and the fitted exponent is a confounded average of several.

Why the choice of projection does not matter. Every route of Sec. 3 divides by $\chi'(\psi_m) = \prod_{l \neq m} (\psi_m - \psi_l)$: explicitly in (a) and (b), and implicitly in the Cayley–Hamilton

form, whose Vandermonde determinant is $\prod_{i<j}(\psi_i - \psi_j)$. With a cluster of size r containing ψ_m , $\chi'(\psi_m) = O(\delta^{r-1}\Delta^{n-r})$, while the numerators—adjugate, generator contraction, or Vandermonde right-hand side—are $O(\Delta^{n-1})$ with absolute error $O(u\Delta^{n-1})$. Every assembly therefore carries

$$|\delta U|_{\text{assembly}} = O(u g^{-(r-1)}), \quad (25)$$

the same exponent with a different constant. The four projections cannot differ by more than an $O(1)$ factor, and any fifth reading of Eq. (1) will not either. This is the practical content of the unification: the equivalence established in Sec. 3 is not only algebraic but numerical, so there is nothing to be gained by choosing among the forms on accuracy grounds, and nothing to be gained by deriving another.

Propagation to probabilities. Eigenvalue error enters Eq. (1) twice: through the phase, at $O(L\delta\psi)$, and through the projectors, at $O(\delta\psi/\delta)$. Writing $\varphi = \Delta L$ for the accumulated phase,

$$|\delta P| = O\left(u \frac{\varphi g + 1}{g^r}\right), \quad (26)$$

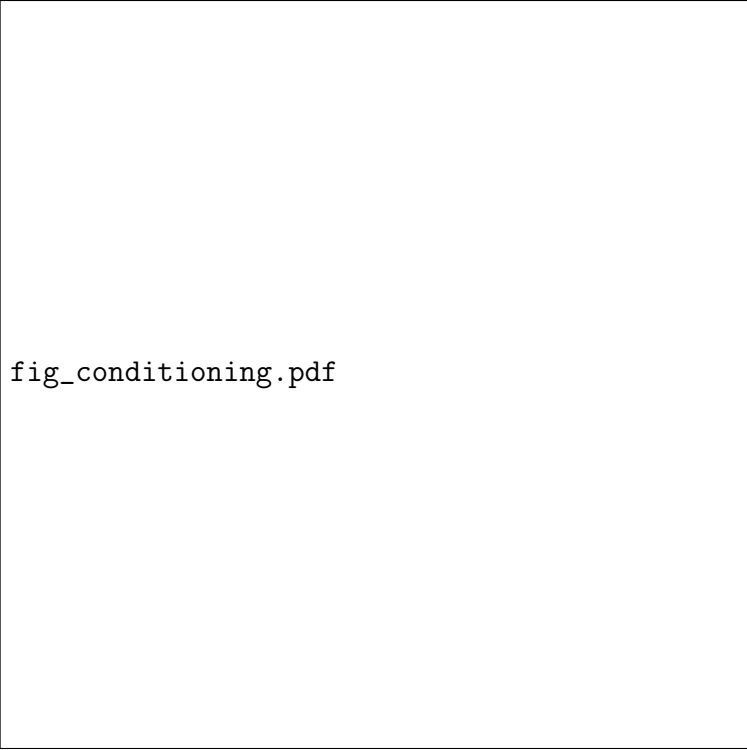
so the error grows linearly in the baseline once $\varphi g \gtrsim 1$, which is the regime of any long-baseline application. The linear-in- L growth is therefore not an empirical observation but a consequence of the roots being wrong: a phase error accumulates, a projector error does not.

The exception. Projection (c) escapes the exponent. By Cauchy interlacing the eigenvalues $\mu_k^{(\alpha)}$ of the principal minor M_α interlace the ψ_l , so a cluster of size r traps $r - 1$ of the $\mu_k^{(\alpha)}$ inside it. The numerator of Eq. (8) then acquires exactly the $r - 1$ vanishing factors that the denominator has, and the near-singularity is removable identically, for every n and every r . What survives in floating point is the relative error of the small differences themselves, so

$$|\delta|V_{\alpha m}|^2 = O(u/g), \quad (27)$$

independent of the cluster size. The eigenvector–eigenvalue identity is thus better conditioned than every phase-carrying route, by a factor $O(\varphi g + 1)g^{-(r-2)}$ —and better than a backward-stable eigensolver on the same object, since the interlacing cancellation is exact where the eigensolver’s is not. It is, however, only better and not immune: one power of $1/g$ remains. Note also that (c) returns moduli alone; whether the advantage survives the phase reconstruction of (d) is not settled here.

Why the obvious repairs fail. Equations (24) and (25) locate the loss in the extraction of the roots, not in the evaluation that follows, so any remedy acting on the evaluation is addressing the wrong step. Two are worth naming because they are the standard recommendations. Rewriting the Vandermonde solve in Newton form with divided differences evaluated through Opitz’s theorem—continuous through confluence by construction [36]—helps at short baselines, where the φg term of Eq. (26) is subdominant, and is indistinguishable from the monomial form once it is not. Deflating the dominant root and solving the residual cubic in closed form is worse than not deflating, since deflation perturbs the remaining



fig_conditioning.pdf

Figure 4: **Where the gap is small.** The relative spectral gap g of Eq. (23) over the 3+1 plane, in constant crust matter. This is a property of $\mathbf{H}$: no evaluation route, arithmetic or machine enters. The error of any route follows from g and the cluster size through Eqs. (24)–(26).

coefficients and Eq. (24) applies again to a spectrum whose spread has been reduced. No reformulation that still passes through χ can evade the exponent. That is the useful result of this section, and it is a negative one.

Where the gap is small. The exponents above are properties of the arithmetic; where they bite is a property of the Hamiltonian. Figure 4 maps g over the 3+1 plane, and involves no evaluation route at all. Combined with Eqs. (24) and (26) it gives the error anywhere on the plane for any implementation, which a table of measured errors would not. The small-gap band is not a measure-zero set; it is a broad ridge crossing the region of experimental interest, and g falls to $\sim 10^{-5}$ within it. At $n = 4$, where the exponent reaches two, that is enough for Eq. (26) to reach the level of the DMP approximation’s own accuracy at long baseline—which is to say, enough to matter for a fit. This is the one place we attach a scale to the asymptotics, and it is the only one needed.

A switching criterion. Both g and Δ follow from the invariants at negligible cost, and the cluster size follows from the sorted eigenvalue differences. A code can therefore evaluate the right-hand side of Eq. (26), compare it against its own tolerance and fall back on an eigensolver—the recommendation of Sec. 10. We give the criterion as a scaling and not as a calibrated bound deliberately: the constants depend on the library, on floating-point

contraction and on the order of operations, and would not transfer, whereas the exponents are properties of the spectrum.

Scope and an outstanding check. Equations (24) through (27) assume only that $\tilde{\mathbf{H}}$ is Hermitian and that the coefficients of χ are formed to relative accuracy $O(u)$; nothing in them is specific to matter, to 3+1, or to the standard interaction. Vacuum, non-standard interactions and antineutrinos change g and the cluster structure, and therefore the location of the ridge in Fig. 4, but not the exponents. What has *not* been done is an independent cross-check against another implementation—**nuSQuIDS**, or the published closed forms of Refs. [9, 10] at a common parameter point. The asymptotics above do not require one; any absolute statement about a particular code would.

At $n = 4$ the closed form is neither faster nor more accurate than a good eigensolver, and Eq. (25) says no rearrangement of it will be. Its advantages are structural, and one survives scrutiny: differentiability.

6.3 Exact gradients

Fits need $\partial P/\partial\theta$. Differentiating an iterative eigensolver is awkward and differentiating eigenvectors is ill-defined at degeneracies, but neither is necessary: the eigenvalues are roots of a polynomial whose coefficients are polynomial in $\mathbf{H}$, so implicit differentiation gives them exactly,

$$\frac{\partial\psi_m}{\partial\theta} = -\frac{1}{\chi'(\psi_m)} \left[\frac{\partial c_2}{\partial\theta} \psi_m^2 + \frac{\partial c_3}{\partial\theta} \psi_m + \frac{\partial c_4}{\partial\theta} \right], \quad \frac{\partial I_2}{\partial\theta} = \text{Tr}(\tilde{\mathbf{H}} \partial_\theta \tilde{\mathbf{H}}), \dots \quad (28)$$

with the remaining derivatives following from App. B. The coefficients c_j of the polynomial form then differentiate through the Vandermonde solve, and U through the product rule. No eigenvector is ever differentiated, and the whole chain is a fixed sequence of arithmetic operations—what automatic differentiation wants, and it ports unchanged to JAX or to a GPU, where the eigenvalue step becomes a custom root rule and everything downstream differentiates itself. The technique is how lattice gauge theory computes hybrid Monte Carlo forces [37–39]; it appears unused in neutrino physics.

The conditioning of the two routes differs by one power of the gap. Standard perturbation theory gives $\partial v_m/\partial\theta = \sum_{l \neq m} v_l \langle v_l | \partial_\theta \tilde{\mathbf{H}} | v_m \rangle / (\psi_m - \psi_l)$, so differentiating eigenvectors introduces a further factor $1/\delta$ on top of whatever the eigenvectors themselves cost: with a cluster of size r the derivative carries $O(u g^{-r})$, and it is undefined in the limit. Implicit differentiation of χ divides by $\chi'(\psi_m)$ once and no more, so by Eq. (25) it carries $O(u g^{-(r-1)})$ —the same conditioning as the probability itself, and one power better than differentiating the eigenvectors. That gap, not a benchmark timing, is the argument for the algebraic route. Figure 5 confirms the construction against finite differences of an arbitrary-precision reference: the residual tracks the reference’s own $1/h$ error rather than the analytic gradient’s, which is the signature of a derivative limited by the comparison and not by itself.

The derivative that matters most. Nothing privileges mixing parameters. The same implicit differentiation delivers $\partial P/\partial\rho$, because the matter potential enters $\mathbf{H}$ linearly and

$\partial\tilde{\mathbf{H}}/\partial\rho$ is a constant diagonal matrix, with the same conditioning as the mixing derivatives. This deserves emphasis because it is the derivative Earth-tomography analyses actually need and the one finite differences handle worst: each evaluation re-solves the full profile, and the step must be tuned against a probability that oscillates rapidly in ρ . Along a segmented profile the chain rule gives $\partial P/\partial\rho_k$ for every slab at the cost of one extra matrix multiplication per slab, so a gradient-based fit to a profile with dozens of layers costs about as much as one forward evaluation. We are not aware of an oscillation code that exposes this.

Note that $\chi'(\psi_m)$ reappears in the denominator of Eq. (28): the conditioning map transfers verbatim to the gradients, and is worst at the same level crossings. Differentiability is a real advantage of the algebraic route, but it is not a way around its conditioning.

7 Adjacent territory

7.1 Approximation schemes and where they attach

The exact construction isolates all of the difficulty into one place—the eigenvalues—and much of the literature on approximate oscillation probabilities can be read as strategies for that one step. Schemes that expand the *probability* directly in $\alpha = \Delta m_{21}^2/\Delta m_{31}^2$ and s_{13} produce the familiar three-term appearance formulas [40–44] and diverge as the atmospheric resonance is approached, because the underlying expansion has a vanishing eigenvalue denominator there. Schemes that instead approximate the *eigenvalues* and feed them into the exact machinery—DMP [24, 25] and NuFast [45]—inherit the resonance behaviour of the exact solution and are uniformly accurate. The systematic comparison is Ref. [46].

In the language of Eq. (5), the divergence of the first family is the statement that $\chi'(\psi_m) \rightarrow 0$ has been expanded rather than resummed. The DMP rotation removes the near-degeneracy exactly before expanding, so the unbounded resonant denominator is absorbed into a matter-dressed angle. NuFast completes the logic: approximate eigenvalues as a seed, one Newton step, then exact assembly. No comparable catalogue exists for four flavours, and its absence is the most useful piece of missing work in the area.

7.2 Matter-potential invariants

Projection (b) has a physics output the other projections obscure: quantities built from eigenvalues and eigenvector bilinears that are independent of the matter potential. The three-flavour set is classical. The Jarlskog invariant [47] is rescaled in matter by the Naumov–Harrison–Scott identity [21, 22],

$$J_{\text{mat}} \Delta\lambda_{21}\Delta\lambda_{31}\Delta\lambda_{32} = J \Delta m_{21}^2 \Delta m_{31}^2 \Delta m_{32}^2, \quad (29)$$

which follows from the matter independence of $\Im(\mathbf{H}_{e\mu}\mathbf{H}_{\mu\tau}\mathbf{H}_{\tau e})$; the KTY identity [4] plays the same role for $|V_{e1}V_{e2}V_{e3}|^2$, and their ratio is the Toshev identity [23]. Convention subtleties in taking the square root are discussed in Ref. [48].

Abdullahi and Parke [13] generalized the construction to $n \geq 4$ from the matter inde-

pendence of $\sum_{\gamma \neq \alpha, \beta} \Im(\mathbf{H}_{\alpha\beta} \mathbf{H}_{\beta\gamma} \mathbf{H}_{\gamma\alpha})$, obtaining

$$\sum_{i>j} \Delta m_{ij}^2 \Delta m_{jk}^2 \Delta m_{ki}^2 J_{ij}^{\alpha\beta}(U) = \sum_{i>j} \Delta \lambda_{ij} \Delta \lambda_{jk} \Delta \lambda_{ki} J_{ij}^{\alpha\beta}(V) \quad (30)$$

for any fixed k , with $(n-1)(n-2)/2$ independent relations—the number of Dirac phases. Off-diagonal non-standard interactions break these identities, which is itself a diagnostic. These invariants are why projection (b) survives despite being less economical than (a): they are invisible in the (u_0, u_a) variables. They are also the strongest available test of an implementation, and Sec. 6 uses them as such.

7.3 Beyond-Standard-Model Hamiltonians

The practical argument for a general- $\mathbf{H}$ interface is that the interesting Hamiltonians are not the standard one. Sterile states, non-standard interactions ($V \rightarrow \sqrt{2}G_F n_e(\mathbf{1} + \epsilon)$ with ϵ Hermitian) [13], scalar NSI acting on the mass matrix [49], Lorentz-invariance violation, and non-unitarity recast as a closed system with extra states are all Hermitian and constant in a slab, hence covered verbatim by Eq. (5). In each case the only quantities that change are h_a and hence the invariants; the eigenvalue solver, the star tower and the probability assembly are untouched.

7.4 Varying density

Constant density is excellent for long-baseline crust chords [50] and inadequate for solar, supernova and core-crossing atmospheric neutrinos. Three strategies exist and all reduce to the machinery above. *Slab products*, $U = \prod_k \exp(-i\mathbf{H}_k L_k)$ with each factor evaluated exactly, form the first-order Magnus scheme, with error controlled by the neglected commutators. *Magnus integrators* keep those commutators and retain unitarity at every order; because each truncation remains in $\mathfrak{su}(n)$, Eq. (5) supplies the exponential map exactly, so a Magnus integrator at any order costs one closed-form exponential per step. Casas *et al.* [33] report speedups of up to two orders of magnitude over general-purpose ODE solvers. *Adiabatic and Landau-Zener treatments* apply when the density varies slowly on the scale of the oscillation length, with the eigenvalue machinery supplying the instantaneous eigenvalues.

7.5 Non-Hermitian and open-system evolution

Two generalizations are often conflated. *Non-Hermitian* $\mathbf{H}$ describes decay: $\mathbf{H} \rightarrow \mathbf{H} - \frac{i}{2}\Gamma$. Sylvester’s formula does not require Hermiticity, so Eq. (1) survives formally; what fails is everything downstream that used reality of the spectrum—the trigonometric root formulas, the non-negativity of the resolvent-cubic roots at $n = 4$, and the identification of $|V_{\alpha i}|^2$ with probabilities. Three-flavour treatments are given in Refs. [51, 52]; the $n = 4$ case is open. *Genuinely open systems* require the density matrix and a Lindblad term [53, 54], which is projection (e) with non-unitary generators added—what SQuIDS/nuSQuIDS integrate, with the closed-form kernel supplying the unitary part of each step.

8 Beyond $n = 4$

At $n \geq 5$ Abel–Ruffini removes the closed-form-in-radicals eigenvalues. Three remarks calibrate what is lost.

Closed form survives, uselessly. The general quintic is solvable beyond radicals—via the Bring radical and elliptic modular functions (Hermite), or hypergeometric functions (Klein)—but those special functions are themselves evaluated iteratively, so nothing is gained over Newton’s method applied to χ . The radicals were never the point; the eigenvalue-only structure was.

The practical scheme. Seed the eigenvalues—by a DMP-style approximation [24], or by the closed-form $n \leq 4$ solution of a decoupled sub-block when a hierarchy permits—and polish with Newton or Halley. The Sylvester assembly of Eq. (1) then proceeds unchanged. This is in effect what `nuSQuIDS` does within its density-matrix evolution [26]. Explicit closed forms already exist at fixed n : Refs. [35, 55?, 56] give $SU(n)$ exponentials in trace invariants through $n = 5$, and Ref. [?] extends the programme to $SO(n)$ up to $n = 9$ and to G_2 , in each case reducing the matrix powers by Cayley–Hamilton and solving a low-degree polynomial in a few invariants.

Differentiability survives without radicals. Equation (28) requires only that ψ_m be a root of χ , not that it be available in radicals, so a differentiable pipeline is available at every n . The conditioning warnings transfer with it.

Appendix G carries this out for $n = 5$ and $n = 6$: roots of χ from a companion-matrix solve followed by Newton polishing, fed into Eq. (1) with the clustering of Sec. 2. Over 300 random Hermitian matrices at each size the probabilities agree with a direct eigendecomposition, degrading with n exactly as Eq. (24) requires: the exponent is set by the cluster size, which can reach $n - 1$, so the worst case loses one further power of g per added flavour while the eigensolver stays at $O(u\Delta)$. The assertion that everything structural continues past $n = 4$ is therefore not a hope, and the conditioning statement extends without further measurement.

9 Origins, and technique not yet borrowed

The construction has been built more than once, in fields that do not read each other. What follows surveys those origins for a practical reason rather than a historical one: several of these communities carried it past the point where neutrino physics stopped, and Sec. 9.2 collects what is available to take.

Two limits on the survey should be stated first. The instances below are notable rather than exhaustive, and they are the ones a neutrino physicist finds, so the neutrino side of this record is by some margin its best-covered part; no claim is made that the coverage is even. And “independent” here means only that the derivations share no cited source. Absence of citation is good evidence of independence, but it does not exclude transmission through textbooks, colleagues or folklore—channels that leave no bibliographic trace, and one of which is visible in Ohlsson and Snellman’s acknowledgment of a mathematical colleague in their own department.

9.1 Independent origins

Applied numerical linear algebra (from 1966). Putzer [18] gave an eigenvalues-only recursion for e^{At} avoiding the Jordan form; Moler and Van Loan [15, 57] catalogued the whole family—series, ODE, polynomial (Cayley–Hamilton, Lagrange, Newton, Vandermonde), matrix-decomposition, splitting—with the comparative conditioning analysis that is that paper’s main point; and Harris, Fillmore and Smith [14] stated in *SIAM Review* that the exponential and the spectral decomposition follow from nothing but the eigenvalues and the Cayley–Hamilton theorem. Higham [36] is the standard modern account.

Magnetic resonance and quantum control (from 1968). What Sec. 7.4 calls the Magnus expansion is *average Hamiltonian theory* in magnetic resonance, introduced by Waugh, Huber and Haeberlen [58] in 1968, reaching neutrinos only with Ref. [31], and developed there into a design discipline [59]: pulse sequences are engineered so that specified orders of the Magnus series vanish identically, with symmetry arguments cancelling whole classes of higher-order terms. The varying-density neutrino problem is structurally the same, and the transfer of technique appears not to have been attempted.

Multilevel quantum optics (from 1981). Hioe and Eberly [19] introduced the $SU(N)$ coherence vector: Hamiltonian and density matrix expanded in generalized Gell-Mann matrices, evolution as a rotation of a real $(N^2 - 1)$ -vector, and nonlinear constants of motion that are the invariants of Sec. 2.2. Aravind [20] solved the $SU(3)$ case by Cayley–Hamilton for arbitrary initial conditions and extended it to relaxation: projection (a) together with its open-system generalization. Dattoli, Richetta and Torre [60] gave matrix representations of the $SU(3)$ evolution operator and observed that three-level evolution is governed by a matrix equivalent to the Kobayashi–Maskawa matrix—one step from the neutrino problem, and a connection still unexploited.

Lattice gauge theory (from 2004). $SU(3)$ simulations need $\exp(iQ)$ for traceless Hermitian 3×3 Q several times per site per update. Morningstar and Peardon [37] use Cayley–Hamilton, writing $Q^3 - c_1 Q - c_0 \mathbf{1} = 0$ with $c_1 = \frac{1}{2} \text{Tr } Q^2$, $c_0 = \frac{1}{3} \text{Tr } Q^3$ —Eq. (39) with $c_1 = I_2$, $c_0 = \frac{2}{3} I_3$ —and impose $27c_0^2 - 4c_1^3 \leq 0$, the reality condition of Sec. 2.2. They treat the near-degenerate case explicitly. Lüscher [38] works out the same representation with emphasis on regularity and gives a numerically stable algorithm for the exponential and its derivatives, required because hybrid Monte Carlo needs the force. The construction has been extended to $SU(N)$ power series and their differentials [39]. The conditioning question of Sec. 6 and the differentiability programme of Sec. 6.3 are therefore, at $n = 3$, standard lattice practice since 2004 and 2009.

Collective neutrino oscillations (from 1988). The dense-neutrino-gas literature is built natively on the coherence vector. Flavour evolution is written as the precession of a real polarization vector, with an explicit “magnetic spin” and “gyromagnetic ratio”, and the three-flavour case proceeds by replacing the Pauli matrices with the Gell-Mann matrices—the construction of Sec. 2.2. The standard review [?] attributes that generalization to Kim,

Kim and Sze [?] and to Dasgupta and Dighe [?], both neutrino papers; its bibliography cites neither Hioe and Eberly nor Aravind, and neither Magnus nor Waugh. The two-flavour “flavour isospin” picture used throughout, complete with the magnetic-spin analogy, is the Feynman–Vernon–Hellwarth construction of 1957 under a different name.

Neutrino phenomenology (from 1999). Ohlsson and Snellman [5, 6] built the construction from scratch. Their $U(L)$ comes from the Cayley–Hamilton polynomial ansatz with the coefficients fixed by an explicitly inverted Vandermonde system in the eigenvalues, then regrouped into Lagrange–Sylvester form—projection (d) followed by (a), not the $\mathfrak{su}(3)$ expansion, which enters the neutrino literature later with Ref. [7]. They also state that the formalism adapts to any number of flavours but that for five or more the eigenvalues must be found numerically, “since only characteristic equations up to degree four are analytically solvable”. Section 4.3 carries out the four-flavour case they identified as reachable. One consequence of reading the 1999 papers closely is worth recording: the Vandermonde coefficient system was described as previously unwritten in the neutrino literature as recently as 2026 [61], whereas it is displayed in full in Ref. [5], and earlier still in Ref. [57].

9.2 What has not transferred

The interesting question is not who cited whom. It is what each of these communities knows that the others have had to work out again, and what is still sitting unused.

Three things stand out. Lattice gauge theory has already solved our conditioning problem at $n = 3$: Morningstar and Peardon treat the near-degenerate case explicitly, and Lüscher gives a numerically stable algorithm for the $SU(3)$ exponential together with its derivatives, because hybrid Monte Carlo cannot function without exact forces. Both the failure mode of Sec. 6 and the gradient construction of Sec. 6.3 are, at $n = 3$, standard practice there. Adapting Lüscher’s algorithm to $n = 4$ is a concrete piece of work that would improve on anything in this paper.

Magnetic resonance has a design discipline we do not. Average Hamiltonian theory is not merely the Magnus expansion under another name; it is fifty years of technique for *engineering* the expansion—choosing pulse sequences so that specified orders vanish identically, and using symmetry to cancel whole classes of higher-order terms. The varying-density problem of Sec. 7.4 is structurally the same: a piecewise-controlled Hamiltonian whose truncation error one wants to suppress. Whether those cancellation arguments carry over to Earth-density profiles is, as far as we know, untested.

And the closed-form $SU(n)$ exponential is a solved problem in mathematical physics well beyond $n = 4$. Van Kortryk gives $SU(4)$ and $SU(5)$ in trace invariants; Kaiser adds $Sp(2)$ and sketches $SU(5)$, $SU(6)$ and $Sp(3)$, and has since carried the same programme to $SO(n)$ up to $n = 9$ and to G_2 . Anyone contemplating five-flavour phenomenology should start there rather than from Sec. 8.

The reason these transfers have not happened is simply that the literatures do not meet: the coherence-vector and average-Hamiltonian work is, as far as we can determine, unknown in neutrino physics, and what little has crossed over came in through the Magnus and geometric-integration channel, carried by numerical analysts moving into the field rather

than sought out from within it. The practical consequence is the point: there is usable technique in at least three neighbouring fields that this one has not picked up.

10 What to use

Situation	Use	Why
Three flavors, production LBL fit	NuFast-LBL	Fastest, 10^{-9} accurate; nothing here beats it
Any n , single evaluation, accuracy paramount	backward-stable eigensolver	$O(u\Delta)$ error, independent of the gap (Eq. (24))
Large energy grids, $n \leq 4$, away from crossings	closed form, any projection	Branch-free, vectorizes; all four agree to $O(1)$ (Eq. (25))
Gradient-based fitting, any n	closed form + implicit differentiation, Eq. (28)	The one advantage that survives; no eigenvector differentiation
Near a level crossing or resonance	eigensolver (<i>not</i> divided differences)	Closed form pays $g^{-(r-1)}$; the error is in the roots, so a better polynomial evaluation does not help (Sec. 6.2)
Varying density	Magnus/slab with closed-form exponential map	Unitary at every order
Open systems, interactions, $n \leq 6$	nuSQuIDS	Only option; closed form does not apply
Analytic insight (matter angles, invariants)	adjugate/EEI route (b)–(d)	The (u_0, u_a) variables hide this structure

Table 3: Recommendations. The honest summary is that the algebraic route is rarely the fastest and never the most accurate; it is worth using when one needs differentiability, vectorization, or an arbitrary Hamiltonian.

11 Open problems

1. *A better root, not a better evaluation.* Section 6 establishes where the closed form fails and shows that divided differences do not repair it. A deflation scheme that resolves the small roots accurately, or the lattice algorithm of Ref. [38] adapted to $n = 4$, would.
2. *Differentiable probabilities in a public code.* Section 6.3 demonstrates the mechanism; no public neutrino code exposes it. A JAX implementation with the custom root rule, benchmarked on a gradient-based 3+1 fit, is the obvious next step.
3. *Four-flavour approximation schemes.* There is no $n = 4$ analogue of the DMP rotation or of the accuracy catalogue of Ref. [46].

4. *Magnus* $\circ$ $SU(n)$ at $n = 4$. A Lie-group integrator using Eq. (5) as the exact exponential map for varying-density 3+1 propagation composes standard ingredients that have not been composed—and the pulse-sequence design techniques of average Hamiltonian theory have never been tried on it.
5. *Non-Hermitian* $n = 4$. Decay and absorption break the reality arguments that make Eq. (19) work.

12 Conclusions

There is one construction here, not many: the resolvent evaluated at its poles. The methods populating the neutrino literature are its projections, and the choice among them is a choice of which information to keep—amplitudes, eigenvectors, or moduli—and in which basis, traded against conditioning and against interpretability. The $SU(4)$ case completes the closed-form programme: at $n = 4$ the characteristic quartic still yields, through a resolvent cubic that reuses the $SU(3)$ trigonometric formula, and beyond $n = 4$ the radicals stop while everything structural continues.

The practical finding is that the closed form should be used for what it is good at. It is neither the fastest nor the most accurate route at $n = 4$; it is branch-free, vectorizable, takes an arbitrary Hermitian Hamiltonian, and—the one advantage that survives scrutiny—differentiates exactly. Near level crossings it pays a factor $g^{-(r-1)}$ in the relative spectral gap, with r the number of eigenvalues that cluster, and the loss is not repairable within the closed form: it is the classical instability of extracting eigenvalues from a characteristic polynomial, so neither divided differences nor deflation nor better coefficient arithmetic recovers it, and the only fix is to stop using χ . Equation (26) says when.

The apparatus also has origins outside this field—in applied mathematics, in magnetic resonance, in quantum optics, and in lattice gauge theory—and in the first three of those the neutrino version arrived last. That ordering is of little interest as history. It matters as a map of where to look next: the conditioning problem of Sec. 6 has a published solution at $n = 3$ in the lattice literature, the varying-density problem of Sec. 7.4 has fifty years of pulse-sequence technique waiting in magnetic resonance, and the five-flavour case is already solved as an algebra problem. If this paper has a function beyond the four-flavour extension, it is to point at those three doors.

Reproducibility

Every figure in Sec. 6 is produced by a single driver script in the accompanying repository. The claims of Sec. 6.2 are asymptotic in u and g and carry no machine-dependent content; the figures that verify them report errors normalised by $u\Delta$, referred to an `mpmath` reference at 50–60 decimal digits. Constants—and therefore any absolute error quoted for a particular code—depend on the linear-algebra library, on floating-point contraction and on the order of operations, which is why none is quoted here.

Code availability

The $SU(4)$ reference implementation provides the generators, invariants, closed-form eigenvalues, both projections, batched implementations, the 3+1 matter Hamiltonian, the validation suite, the conditioning study, the Opitz implementation, and the gradient machinery. NumPy is the only dependency. The bibliographic scripts used for Sec. 9.2 are included for completeness.

A Generators and conventions

The $n^2 - 1$ generalized Gell-Mann matrices are: for each pair $j < k$, the symmetric $S^{(jk)}$ with $S_{jk}^{(jk)} = S_{kj}^{(jk)} = 1$ and the antisymmetric $A^{(jk)}$ with $A_{jk}^{(jk)} = -i$, $A_{kj}^{(jk)} = +i$ [$n(n-1)$ of them]; and the $n-1$ Cartan matrices

$$D_l = \sqrt{\frac{2}{l(l+1)}} \text{diag}(\underbrace{1, \dots, 1}_l, -l, 0, \dots, 0), \quad l = 1, \dots, n-1. \quad (31)$$

All satisfy $\text{Tr}(\lambda^a \lambda^b) = 2\delta_{ab}$ and

$$\lambda^a \lambda^b = \frac{2}{n} \delta_{ab} \mathbf{1} + (d_{abc} + i f_{abc}) \lambda^c, \quad d_{abc} = \frac{1}{4} \text{Tr}(\{\lambda^a, \lambda^b\} \lambda^c), \quad f_{abc} = \frac{1}{4i} \text{Tr}([\lambda^a, \lambda^b] \lambda^c), \quad (32)$$

with d totally symmetric and f totally antisymmetric. For $n=2$ the d tensor vanishes and Eq. (32) is the Pauli identity; for $n=3$ the ordering used in Table 2 is the standard one ($\lambda^{1,2}$ on the $(1,2)$ pair, $\lambda^3 = D_1$, $\lambda^{4,5}$ on $(1,3)$, $\lambda^{6,7}$ on $(2,3)$, $\lambda^8 = D_2$). For $n=4$ we place the six pairs first, in lexicographic order and symmetric-before-antisymmetric, and the three Cartan generators last, as stated in Table 2; nothing depends on this choice except the index labels.

B Powers of $\tilde{\mathbf{H}}$, the star tower, and the $SU(3)$ accident

Write $\tilde{\mathbf{H}} = h_a \lambda^a$. Using Eq. (32),

$$\tilde{\mathbf{H}}^2 = h_a h_b \lambda^a \lambda^b = \frac{2}{n} |\mathbf{h}|^2 \mathbf{1} + (d_{abc} + i f_{abc}) h^a h^b \lambda^c = \frac{2}{n} I_2 \mathbf{1} + (\mathbf{h} * \mathbf{h})_c \lambda^c, \quad (33)$$

the f term vanishing because $h^a h^b$ is symmetric. Multiplying once more,

$$\begin{aligned} \tilde{\mathbf{H}}^3 &= \tilde{\mathbf{H}} \left[\frac{2}{n} I_2 \mathbf{1} + (\mathbf{h} * \mathbf{h})_a \lambda^a \right] = \frac{2}{n} I_2 h_c \lambda^c + h_b (\mathbf{h} * \mathbf{h})_a \lambda^b \lambda^a \\ &= \frac{2}{n} I_3 \mathbf{1} + \left[\frac{2}{n} I_2 h_c + ((\mathbf{h} * \mathbf{h}) * \mathbf{h})_c \right] \lambda^c, \end{aligned} \quad (34)$$

where $I_3 = d_{abc} h^a h^b h^c = \mathbf{h} \cdot (\mathbf{h} * \mathbf{h})$ and the f contraction $f_{abc} h^b (\mathbf{h} * \mathbf{h})^a$ vanishes—equivalently, $[\tilde{\mathbf{H}}, \tilde{\mathbf{H}}^2] = 0$. Squaring Eq. (33),

$$\tilde{\mathbf{H}}^4 = \frac{4}{n^2} I_2^2 \mathbf{1} + \frac{4}{n} I_2 (\mathbf{h} * \mathbf{h})_a \lambda^a + (\mathbf{h} * \mathbf{h})_a (\mathbf{h} * \mathbf{h})_b \lambda^a \lambda^b. \quad (35)$$

Taking traces of Eqs. (33)–(35) and using $\text{Tr } \lambda^a = 0$ gives the invariants used throughout:

$$\text{Tr } \tilde{\mathbf{H}}^2 = 2I_2, \quad \text{Tr } \tilde{\mathbf{H}}^3 = 2I_3, \quad \text{Tr } \tilde{\mathbf{H}}^4 = \frac{4}{n}I_2^2 + 2I_4, \quad I_4 \equiv |\mathbf{h} * \mathbf{h}|^2, \quad (36)$$

which at $n = 4$ is Eq. (14). Projecting Eqs. (33)–(34) with $\frac{1}{2} \text{Tr}(\lambda^c \cdot)$ gives the star tower without ever constructing d_{abc} —the practical point that makes the method cheap to implement.

Why the tower truncates at $n = 3$ and not at $n = 4$. For traceless $\tilde{\mathbf{H}}$ at $n = 3$, Cayley–Hamilton reads $\tilde{\mathbf{H}}^3 = I_2 \tilde{\mathbf{H}} + \frac{2}{3} I_3 \mathbf{1}$ (App. C). Comparing with Eq. (34) at $n = 3$ and matching the coefficient of λ^c ,

$$I_2 h_c = \frac{2}{3} I_2 h_c + ((\mathbf{h} * \mathbf{h}) * \mathbf{h})_c \implies (\mathbf{h} * \mathbf{h}) * \mathbf{h} = \frac{1}{3} |\mathbf{h}|^2 \mathbf{h}, \quad (37)$$

so the $\text{SU}(3)$ identity is not an independent fact about the d tensor but a consequence of the characteristic equation at $n = 3$. At $n = 4$ Cayley–Hamilton instead relates $\tilde{\mathbf{H}}^4$ to lower powers, which through Eq. (35) constrains $(\mathbf{h} * \mathbf{h}) * (\mathbf{h} * \mathbf{h})$ and leaves $(\mathbf{h} * \mathbf{h}) * \mathbf{h}$ unconstrained. The tower therefore has $n - 1$ rungs in general, and the numerator $N_a(\psi)$ of Eq. (5) has degree $n - 2$.

C Characteristic polynomials from Newton’s identities

For traceless $\tilde{\mathbf{H}}$ write $\chi(\psi) = \psi^n + \sum_{k \geq 2} (-1)^k e_k \psi^{n-k}$ with e_k the elementary symmetric polynomials of the eigenvalues and $p_k = \text{Tr } \tilde{\mathbf{H}}^k$. Newton’s identities with $p_1 = e_1 = 0$ give $e_2 = -\frac{1}{2} p_2$, $e_3 = \frac{1}{3} p_3$, $e_4 = \frac{1}{8} p_2^2 - \frac{1}{4} p_4$. Substituting Eq. (36):

$$n = 2 : \quad \chi(\psi) = \psi^2 - I_2, \quad (38)$$

$$n = 3 : \quad \chi(\psi) = \psi^3 - I_2 \psi - \frac{2}{3} I_3, \quad (39)$$

$$n = 4 : \quad \chi(\psi) = \psi^4 - I_2 \psi^2 - \frac{2}{3} I_3 \psi + \frac{1}{4} (I_2^2 - 2I_4). \quad (40)$$

Equation (39) is the cubic of Ref. [7]; Eq. (40) is Eq. (17). Note that I_4 enters only at $n = 4$: it is the third Casimir, and its absence at $n = 3$ is the second of the two accidents that make the three-flavor case compact.

D The quartic: Euler’s resolvent and its trigonometric solution

Write Eq. (40) as $\psi^4 + p\psi^2 + q\psi + r = 0$ with $p = -I_2$, $q = -\frac{2}{3} I_3$, $r = \frac{1}{4} (I_2^2 - 2I_4)$. Euler’s method seeks the roots as $\psi_m = \frac{1}{2} (\pm \sqrt{z_1} \pm \sqrt{z_2} \pm \sqrt{z_3})$ with an even number of minus signs. Imposing that these four combinations reproduce the elementary symmetric functions yields the resolvent cubic

$$z^3 + 2p z^2 + (p^2 - 4r) z - q^2 = 0 \iff z^3 - 2I_2 z^2 + 2I_4 z - \frac{4}{9} I_3^2 = 0, \quad (41)$$

whose roots are the three pair-sums squared, $z_1 = (\psi_1 + \psi_2)^2$, $z_2 = (\psi_1 + \psi_3)^2$, $z_3 = (\psi_1 + \psi_4)^2$ (using $\sum_m \psi_m = 0$, so $\psi_1 + \psi_2 = -(\psi_3 + \psi_4)$ and the labelling is unambiguous). Because $\tilde{\mathbf{H}}$ is Hermitian all ψ_m are real, hence all z_i are real and non-negative—this is what allows the *same* trigonometric (Cardano–Viète) branch used for Eq. (39) to be reused here, and it is the reason the four-flavor case inherits the three-flavor formula rather than requiring complex intermediate arithmetic. Explicitly, for a depressed cubic $w^3 + Pw + Q = 0$ with $P < 0$ and three real roots, $w_k = 2\sqrt{-P/3} \cos[\frac{1}{3} \arccos(\frac{3Q}{Pt}) + \frac{2\pi k}{3}]$, $t = 2\sqrt{-P/3}$, $k = 0, 1, 2$.

The sign constraint follows from Viète: the product of the four roots is r and their pairwise structure gives $\sqrt{z_1 z_2 z_3} = |q|$ with

$$s_1 s_2 s_3 \sqrt{z_1 z_2 z_3} = -q = \frac{2}{3} I_3, \quad (42)$$

so that of the eight sign assignments only four are admissible, and they are the four roots. In practice one takes $s_i = +1$ and flips a single sign when $q > 0$; two guarded Newton steps on χ then remove the residual error from the square roots, which is the dominant source of loss when two z_i are close (Sec. 6).

E Explicit numerators

Expanding $\text{adj}(\psi \mathbf{1} - \tilde{\mathbf{H}})$ by Faddeev–LeVerrier and projecting with $\frac{1}{2} \text{Tr}(\lambda^a \cdot)$ as in Sec. 2.2 gives, for the three solvable cases,

$$n = 2: \quad N_a(\psi) = h_a, \quad \chi'(\psi) = 2\psi, \quad (43)$$

$$n = 3: \quad N_a(\psi) = \psi h_a + (\mathbf{h} * \mathbf{h})_a, \quad \chi'(\psi) = 3\psi^2 - I_2, \quad (44)$$

$$n = 4: \quad N_a(\psi) = (\psi^2 - \frac{1}{2} I_2) h_a + \psi (\mathbf{h} * \mathbf{h})_a + ((\mathbf{h} * \mathbf{h}) * \mathbf{h})_a, \quad \chi'(\psi) = 4\psi^3 - 2I_2 \psi - \frac{2}{3} I_3. \quad (45)$$

Substituting Eq. (43) into Eq. (5) and using $\psi_{\pm} = \pm |\mathbf{h}|$ reproduces Eq. (11) directly; Eq. (44) gives Eq. (12); Eq. (45) gives Eq. (20). As a consistency check, the identity component of Eq. (4) requires $\text{Tr } P_m = 1$, i.e. that the ψ^{n-1} coefficient of $\text{Tr adj}(\psi \mathbf{1} - \tilde{\mathbf{H}})$ equal n and that $\text{Tr adj}(\psi \mathbf{1} - \tilde{\mathbf{H}}) = \chi'(\psi)$ identically—a one-line check that catches sign errors in N_a and that we recommend as the first unit test of any implementation.

F Explicit SU(4) generators

In the ordering of App. A, with flavour order $(\nu_e, \nu_\mu, \nu_\tau, \nu_s)$: for each pair (j, k) the symmetric generator has 1 in entries (j, k) and (k, j) and the antisymmetric one has $-i$ in (j, k) and $+i$ in (k, j) , giving $\lambda^{1,2}$ on $(1, 2)$, $\lambda^{3,4}$ on $(1, 3)$, $\lambda^{5,6}$ on $(1, 4)$, $\lambda^{7,8}$ on $(2, 3)$, $\lambda^{9,10}$ on $(2, 4)$ and $\lambda^{11,12}$ on $(3, 4)$. The three Cartan generators are

$$\lambda^{13} = \begin{pmatrix} 1 & & & \\ & -1 & & \\ & & 0 & \\ & & & 0 \end{pmatrix}, \quad \lambda^{14} = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 & & & \\ & 1 & & \\ & & -2 & \\ & & & 0 \end{pmatrix}, \quad \lambda^{15} = \frac{1}{\sqrt{6}} \begin{pmatrix} 1 & & & \\ & 1 & & \\ & & 1 & \\ & & & -3 \end{pmatrix}. \quad (46)$$

All fifteen satisfy $\text{Tr}(\lambda^a \lambda^b) = 2\delta_{ab}$. Deleting $\lambda^{5,6,9,10,11,12,15}$ and relabelling leaves exactly the Gell-Mann basis on the upper-left 3×3 block, which is the embedding property used in Sec. 5. The d -tensor is never needed at runtime—the star tower comes from Eq. (16)—but if wanted it follows from $d_{abc} = \frac{1}{4} \text{Tr}(\{\lambda^a, \lambda^b\} \lambda^c)$ applied to the above.

G The construction at $n = 5$ and $n = 6$

Beyond $n = 4$ the eigenvalues are no longer available in radicals, but Eq. (1) needs only their values. We take the coefficients of χ from $\text{Tr} \tilde{\mathbf{H}}^k$, obtain the roots from a companion-matrix solve—a root-finder applied to χ , not an eigensolver applied to $\mathbf{H}$ —polish them with three Newton steps, cluster them with the tolerance of Sec. 2, and assemble U from the Lagrange product over distinct clusters. Over 300 random Hermitian matrices with L drawn uniformly on $[0.1, 20]$: the closed form tracks a direct eigendecomposition to machine precision away from clusters, and degrades with n in accordance with Eq. (24). The degradation relative to $n \leq 4$ is the same characteristic-polynomial instability quantified in Sec. 6, now without even the consolation of a closed form; for production use at these sizes an eigensolver is the right choice, and the demonstration establishes only that the algebraic structure—projectors, star tower, probability table, implicit differentiation—continues unchanged.

References

- [1] V. D. Barger, K. Whisnant, S. Pakvasa, and R. J. N. Phillips. Matter effects on three-neutrino oscillations. *Phys. Rev. D*, 22:2718, 1980. doi: 10.1103/PhysRevD.22.2718.
- [2] H. W. Zaglauer and K. H. Schwarzer. The mixing angles in matter for three generations of neutrinos and the MSW mechanism. *Z. Phys. C*, 40:273, 1988. doi: 10.1007/BF01555889.
- [3] K. Kimura, A. Takamura, and H. Yokomakura. Exact formula of probability and CP violation for neutrino oscillations in matter. *Phys. Lett. B*, 537:86–94, 2002. doi: 10.1016/S0370-2693(02)01907-8.
- [4] K. Kimura, A. Takamura, and H. Yokomakura. Exact formulas and simple CP dependence of neutrino oscillation probabilities in matter with constant density. *Phys. Rev. D*, 66:073005, 2002. doi: 10.1103/PhysRevD.66.073005.
- [5] T. Ohlsson and H. Snellman. Three flavor neutrino oscillations in matter. *J. Math. Phys.*, 41:2768–2788, 2000. doi: 10.1063/1.533270. [Erratum: *J. Math. Phys.* 42, 2345 (2001)].
- [6] T. Ohlsson and H. Snellman. Neutrino oscillations with three flavors in matter of varying density. *Phys. Rev. D*, 60:093007, 1999. doi: 10.1103/PhysRevD.60.093007.
- [7] M. Bustamante. NuOscProbExact: a general-purpose code to compute exact two-flavor and three-flavor neutrino oscillation probabilities. 2019. Code: <https://github.com/mbustama/NuOscProbExact>.

- [8] P. F. Harrison and W. G. Scott. Neutrino matter effect invariants and the observables of neutrino oscillations. *Phys. Lett. B*, 535:229–235, 2002. doi: 10.1016/S0370-2693(02)01753-5.
- [9] Y. Kamo, S. Yajima, Y. Higashida, S.-I. Kubota, S. Tokuo, and J.-I. Ichihara. Four-neutrino oscillations in matter. *Eur. Phys. J. C*, 28:211–216, 2003. doi: 10.1140/epjc/s2003-01151-3. Ref. [89] of arXiv:1904.12391.
- [10] W. Li, J. Ling, F. Xu, and B. Yue. Matter effect of light sterile neutrino: An exact analytical approach. *JHEP*, 10:021, 2018. doi: 10.1007/JHEP10(2018)021.
- [11] P. B. Denton, S. J. Parke, T. Tao, and X. Zhang. Eigenvectors from eigenvalues: A survey of a basic identity in linear algebra. *Bull. Am. Math. Soc.*, 59:31–58, 2022. doi: 10.1090/bull/1722.
- [12] P. B. Denton, S. J. Parke, and X. Zhang. Neutrino oscillations in matter via eigenvalues. *Phys. Rev. D*, 101:093001, 2020. doi: 10.1103/PhysRevD.101.093001. “Eigenvalues: the Rosetta Stone for neutrino oscillations in matter”.
- [13] A. Abdullahi and S. J. Parke. Neutrino oscillations in matter using the adjugate of the Hamiltonian. *Eur. Phys. J. C*, 84, 2024. doi: 10.1140/epjc/s10052-024-12885-5.
- [14] W. A. Harris, Jr., J. P. Fillmore, and D. R. Smith. Matrix exponentials—another approach. *SIAM Rev.*, 43(4):694–706, 2001. doi: 10.1137/S0036144599362406.
- [15] C. B. Moler and C. F. Van Loan. Nineteen dubious ways to compute the exponential of a matrix, twenty-five years later. *SIAM Rev.*, 45:3–49, 2003. doi: 10.1137/S00361445024180.
- [16] V. N. Faddeeva. *Computational Methods of Linear Algebra*. Dover, 1959.
- [17] S.-H. Hou. Classroom note: A simple proof of the Leverrier–Faddeev characteristic polynomial algorithm. *SIAM Rev.*, 40(3):706–709, 1998. doi: 10.1137/S003614459732076X.
- [18] E. J. Putzer. Avoiding the Jordan canonical form in the discussion of linear systems with constant coefficients. *Amer. Math. Monthly*, 73:2–7, 1966. doi: 10.1080/00029890.1966.11970714.
- [19] F. T. Hioe and J. H. Eberly. N -level coherence vector and higher conservation laws in quantum optics and quantum mechanics. *Phys. Rev. Lett.*, 47:838–841, 1981. doi: 10.1103/PhysRevLett.47.838.
- [20] P. K. Aravind. Matrix solution of pseudospin equations for three-level systems. *J. Opt. Soc. Am. B*, 3(7):1025–1032, 1986. doi: 10.1364/JOSAB.3.001025.
- [21] V. A. Naumov. Three neutrino oscillations in matter, CP violation and topological phases. *Int. J. Mod. Phys. D*, 1:379–399, 1992. doi: 10.1142/S0218271892000203.

- [22] P. F. Harrison and W. G. Scott. CP and T violation in neutrino oscillations and invariance of Jarlskog’s determinant to matter effects. *Phys. Lett. B*, 476:349–355, 2000. doi: 10.1016/S0370-2693(00)00153-2.
- [23] S. Toshev. On T violation in matter neutrino oscillations. *Mod. Phys. Lett. A*, 6: 455–460, 1991. doi: 10.1142/S0217732391000464.
- [24] P. B. Denton, H. Minakata, and S. J. Parke. Compact perturbative expressions for neutrino oscillations in matter. *JHEP*, 06:051, 2016. doi: 10.1007/JHEP06(2016)051.
- [25] P. B. Denton, S. J. Parke, and X. Zhang. Rotations versus perturbative expansions for calculating neutrino oscillation probabilities in matter. *Phys. Rev. D*, 98:033001, 2018. doi: 10.1103/PhysRevD.98.033001. Ref. [44] of arXiv:1904.12391.
- [26] C. A. Argüelles Delgado, J. Salvadó, and C. N. Weaver. A simple quantum integro-differential solver (SQuIDS). *Comput. Phys. Commun.*, 196:569–591, 2015. doi: 10.1016/j.cpc.2015.06.022. Code: <https://github.com/arguelles/nuSQuIDS>.
- [27] C. A. Argüelles, J. Salvadó, and C. N. Weaver. nuSQuIDS: A toolbox for neutrino propagation. *Comput. Phys. Commun.*, 277:108346, 2022. doi: 10.1016/j.cpc.2022.108346.
- [28] W. Magnus. On the exponential solution of differential equations for a linear operator. *Commun. Pure Appl. Math.*, 7:649–673, 1954. doi: 10.1002/cpa.3160070404.
- [29] S. Blanes, F. Casas, J. A. Oteo, and J. Ros. The Magnus expansion and some of its applications. *Phys. Rept.*, 470:151–238, 2009. doi: 10.1016/j.physrep.2008.11.001.
- [30] A. N. Ioannisian and A. Yu. Smirnov. Describing neutrino oscillations in matter with Magnus expansion. 2008.
- [31] J. C. D’Olivo and J. A. Oteo. Magnus expansion and the two-neutrino oscillations in matter. *Phys. Rev. D*, 42:256, 1990. doi: 10.1103/PhysRevD.42.256.
- [32] A. Iserles, H. Z. Munthe-Kaas, S. P. Nørsett, and A. Zanna. Lie-group methods. *Acta Numerica*, 9:215–365, 2000. doi: 10.1017/S0962492900002154.
- [33] F. Casas et al. Efficient numerical integration of neutrino oscillations in matter. 2016.
- [34] J. J. Sylvester. On the equation to the secular inequalities in the planetary theory. *Phil. Mag.*, 16:267–269, 1883.
- [35] T. S. Van Kortryk. Matrix exponentials, $SU(N)$ group elements, and real polynomial roots. *J. Math. Phys.*, 57:021701, 2016. doi: 10.1063/1.4938418.
- [36] N. J. Higham. *Functions of Matrices: Theory and Computation*. SIAM, Philadelphia, 2008. doi: 10.1137/1.9780898717778.
- [37] C. Morningstar and M. J. Peardon. Analytic smearing of $SU(3)$ link variables in lattice QCD. *Phys. Rev. D*, 69:054501, 2004. doi: 10.1103/PhysRevD.69.054501.

- [38] M. Lüscher. SU(3) matrix functions, 2009. Technical note, August 2009, revised February 2020; <https://luscher.web.cern.ch/luscher/notes/su3fcts.pdf>.
- [39] F. Knechtli. Evaluating matrix power series with the Cayley–Hamilton theorem. 2024. Lattice 2024 proceedings; VERIFY published form.
- [40] A. Cervera, A. Donini, M. B. Gavela, J. J. Gomez Cadenas, P. Hernandez, O. Mena, and S. Rigolin. Golden measurements at a neutrino factory. *Nucl. Phys. B*, 579:17–55, 2000. doi: 10.1016/S0550-3213(00)00221-2. [Erratum: Nucl. Phys. B 593, 731 (2001)].
- [41] E. K. Akhmedov, R. Johansson, M. Lindner, T. Ohlsson, and T. Schwetz. Series expansions for three flavor neutrino oscillation probabilities in matter. *JHEP*, 04:078, 2004. doi: 10.1088/1126-6708/2004/04/078.
- [42] M. Freund. Analytic approximations for three neutrino oscillation parameters and probabilities in matter. *Phys. Rev. D*, 64:053003, 2001. doi: 10.1103/PhysRevD.64.053003.
- [43] H. Minakata and S. J. Parke. Simple and compact expressions for neutrino oscillation probabilities in matter. *JHEP*, 01:180, 2016. doi: 10.1007/JHEP01(2016)180.
- [44] K. Asano and H. Minakata. Large- θ_{13} perturbation theory of neutrino oscillation for long-baseline experiments. *JHEP*, 06:022, 2011. doi: 10.1007/JHEP06(2011)022.
- [45] P. B. Denton and S. J. Parke. Fast and accurate algorithm for calculating long-baseline neutrino oscillation probabilities with matter effects: NuFast. *Phys. Rev. D*, 110:073005, 2024. doi: 10.1103/PhysRevD.110.073005.
- [46] G. Barenboim, P. B. Denton, S. J. Parke, and C. A. Ternes. Neutrino oscillation probabilities through the looking glass. *Phys. Lett. B*, 791:351–360, 2019. doi: 10.1016/j.physletb.2019.03.002.
- [47] C. Jarlskog. Commutator of the quark mass matrices in the standard electroweak model and a measure of maximal CP nonconservation. *Phys. Rev. Lett.*, 55:1039, 1985. doi: 10.1103/PhysRevLett.55.1039.
- [48] P. B. Denton and S. J. Parke. Parameter symmetries of neutrino oscillations in vacuum, matter, and approximation schemes. *Phys. Rev. D*, 105:013002, 2022. doi: 10.1103/PhysRevD.105.013002.
- [49] A. Medhi et al. Neutrino oscillations in presence of diagonal elements of scalar NSI: an analytic approach. *JHEP*, 12:222, 2024. doi: 10.1007/JHEP12(2024)222. VERIFY full author list before submission.
- [50] K. J. Kelly and S. J. Parke. Matter density profile shape effects at DUNE. *Phys. Rev. D*, 98:015025, 2018. doi: 10.1103/PhysRevD.98.015025.
- [51] D. S. Chattopadhyay, K. Chakraborty, A. Dighe, and S. Goswami. Analytic treatment of 3-flavor neutrino oscillation and decay in matter. *JHEP*, 01:051, 2023. doi: 10.1007/JHEP01(2023)051.

- [52] J. Grönroos, T. Ohlsson, and S. Vihonen. Revisiting series expansions of neutrino oscillation and decay probabilities in matter. *Phys. Rev. D*, 111:035003, 2025. doi: 10.1103/PhysRevD.111.035003.
- [53] G. Lindblad. On the generators of quantum dynamical semigroups. *Commun. Math. Phys.*, 48:119, 1976. doi: 10.1007/BF01608499.
- [54] V. Gorini, A. Kossakowski, and E. C. G. Sudarshan. Completely positive dynamical semigroups of N -level systems. *J. Math. Phys.*, 17:821, 1976. doi: 10.1063/1.522979.
- [55] D. Kusnezov. Exact matrix expansions for group elements of $SU(N)$. *J. Math. Phys.*, 36:898, 1995. doi: 10.1063/1.531165.
- [56] T. L. Curtright and C. K. Zachos. Elementary results for the fundamental representation of $SU(3)$. *Rept. Math. Phys.*, 76:401–404, 2015. doi: 10.1016/S0034-4877(15)30040-9.
- [57] C. B. Moler and C. F. Van Loan. Nineteen dubious ways to compute the exponential of a matrix. *SIAM Rev.*, 20:801–836, 1978. doi: 10.1137/1020098.
- [58] J. S. Waugh, L. M. Huber, and U. Haeberlen. Approach to high-resolution NMR in solids. *Phys. Rev. Lett.*, 20:180, 1968. doi: 10.1103/PhysRevLett.20.180.
- [59] U. Haeberlen. *High Resolution NMR in Solids: Selective Averaging*. Academic Press, 1976.
- [60] G. Dattoli, M. Richetta, and A. Torre. Evolution of $SU(2)$ and $SU(3)$ states: a novel approach through the Cayley–Klein decomposition. *J. Math. Phys.*, 28:618, 1987. See also G. Dattoli, A. Torre, “Matrix representation of the evolution operator for the $SU(3)$ dynamics”; VERIFY volume/pages before submission.
- [61] A. Chaulagain, A. Dhakal, and D. N. Chhatkuli. Algebraic structure of three-flavor neutrino oscillations in constant-density matter: Cayley–Hamilton evolution, DMP resummation, and closed-form uncertainty propagation. 2026.

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Figure 5: **Exact gradients.** Left, $\partial P_{\nu_\mu \rightarrow \nu_s} / \partial \theta_{14}$ from implicit differentiation of χ (solid) against a central finite difference (dashed), at $E = 1$ GeV, $L = 1300$ km, $\Delta m_{41}^2 = 1$ eV². Right, their relative difference. No eigenvector is differentiated at any point.